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MIL-M-28787/519(SH)
13 November 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY INTERFACE I MODULE, KEY CODE CWG

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic memory interface I module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.253 failures per million operating hours for class I or 2.915 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CWG.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1-D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D9	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D10	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D11	Ibd	V	S	0.0	1.4	0.0	1.4	0.0	1.4	mA
D12	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D13	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D14-D16	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D17	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D18	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D19	Ibd	V	S	-0.12	1.21	-0.12	1.21	-0.12	1.21	mA
D20-D26	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D27	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D28-D35	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D36	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D37	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D38	Ibd	V	S	0.0	1.3	0.0	1.3	0.0	1.3	mA
D39	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D40-D48	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E20	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E21	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E22	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E23,E24	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E25-E48	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E49,E50	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA

- See footnotes on next page.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E51-E53	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E54	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E55-E58	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E59-E62	Ios	VI	S	-90.0	-12.0	-90.0	-12.0	-90.0	-12.0	mA
E63	Ios	VI	S	-68.5	-28.8	-68.5	-28.8	-68.5	-28.8	mA
F1-F8	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F9	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F10	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F11	Iih	VII	X	-4.4	0.13	-4.4	0.13	-4.4	0.13	mA
F12	Iih	VII	X	-3.2	0.09	-3.2	0.09	-3.2	0.09	mA
F13	Iih	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
F14-F16	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F17	Iih	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
F18	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F19	Iih	VII	X	-0.03	-2.8	-0.03	-2.8	-0.03	-2.8	mA
F20-F26	Iih	VII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
F27,F28	Iih	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F29	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F30-F37	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F38	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F39	Iih	VII	X	-3.2	0.09	-3.2	0.09	-3.2	0.09	mA
F40	Iih	VII	X	-3.8	0.11	-3.8	0.11	-3.8	0.11	mA
F41	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F42	Iih	VII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
F43,F44	Iih	VII	X	-0.1	0.01	-0.1	0.01	-0.1	0.01	mA
F45-F50	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G1-G8	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G9	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G10	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G11	Iil	VIII	X	-4.4	0.13	-4.4	0.13	-4.4	0.13	mA
G12	Iil	VIII	X	-3.2	0.09	-3.2	0.09	-3.2	0.09	mA
G13	Iil	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G14-G16	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G17	Iil	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
G18	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G19	Iil	VIII	X	-8.03	-2.8	-8.03	-2.8	-8.03	-2.8	mA
G20-G26	Iil	VIII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
G27,G28	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G29	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G30-G37	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G38	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G39	Iil	VIII	X	-3.2	0.09	-3.2	0.09	-3.2	0.09	mA
G40	Iil	VIII	X	-3.8	0.11	-3.8	0.11	-3.8	0.11	mA
G41	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G42	Iil	VIII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
G43,G44	Iil	VIII	X	-0.1	0.01	-0.1	0.01	-0.1	0.01	mA
G45-G50	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
H1-H62	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J65	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1	Iozh	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	mA
K2	Iozh	XI	X	0.0	0.25	0.0	0.25	0.0	0.25	mA
K3	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K4	Iozh	XI	X	0.0	0.25	0.0	0.25	0.0	0.25	mA
K5	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L18	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L19-L25	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L26-L38	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L39	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L40-L47	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	5.7	10.7	5.7	10.7	5.7	10.7	uF
M2	PSFC	XIII	S	5.0	9.4	5.0	9.4	5.0	9.4	uF

- See footnotes at end of table

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
N	Icc	XIV	X	467	1121	467	1121	467	1121	mA
P1,P2	Functional	XV	X	----------		See table XV		----------		
Q1,Q2	Dynamic	XVI	S	----------		See table XVI		----------		
Q3,Q4	Dynamic	XVI	Q	----------		See table XVI		----------		
R1-R8	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R9	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R10	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R11	Ci/o	XVIII	Q	6	125	6	125	6	125	pF
R12	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R13	Ci/o	XVIII	Q	8	150	8	150	8	150	pF
R14	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R15,R16	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R17	Ci/o	XVIII	Q	8	125	8	125	8	125	pF
R18	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R19	Ci/o	XVIII	Q	6	75	6	75	6	75	pF
R20	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R21,R22	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R23-R26	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R27	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R28-R34	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R35	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R36	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R37	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R38	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R39	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R40	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R41,R42	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R43-R48	Ci/o	XVIII	Q	2	50	2	50	2	50	pF

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R49-R56	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R57	C1/o	XVIII	Q	8	125	8	125	8	125	pF
R58	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R59,R60	C1/o	XVIII	Q	8	100	8	100	8	100	pF
R61	C1/o	XVIII	Q	8	125	8	125	8	125	pF
R62	C1/o	XVIII	Q	8	100	8	100	8	100	pF
R63-R66	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R67	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R68,R69	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R70,R71	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R72	C1/o	XVIII	Q	4	100	4	100	4	100	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/519-1 for class I and M28787/519-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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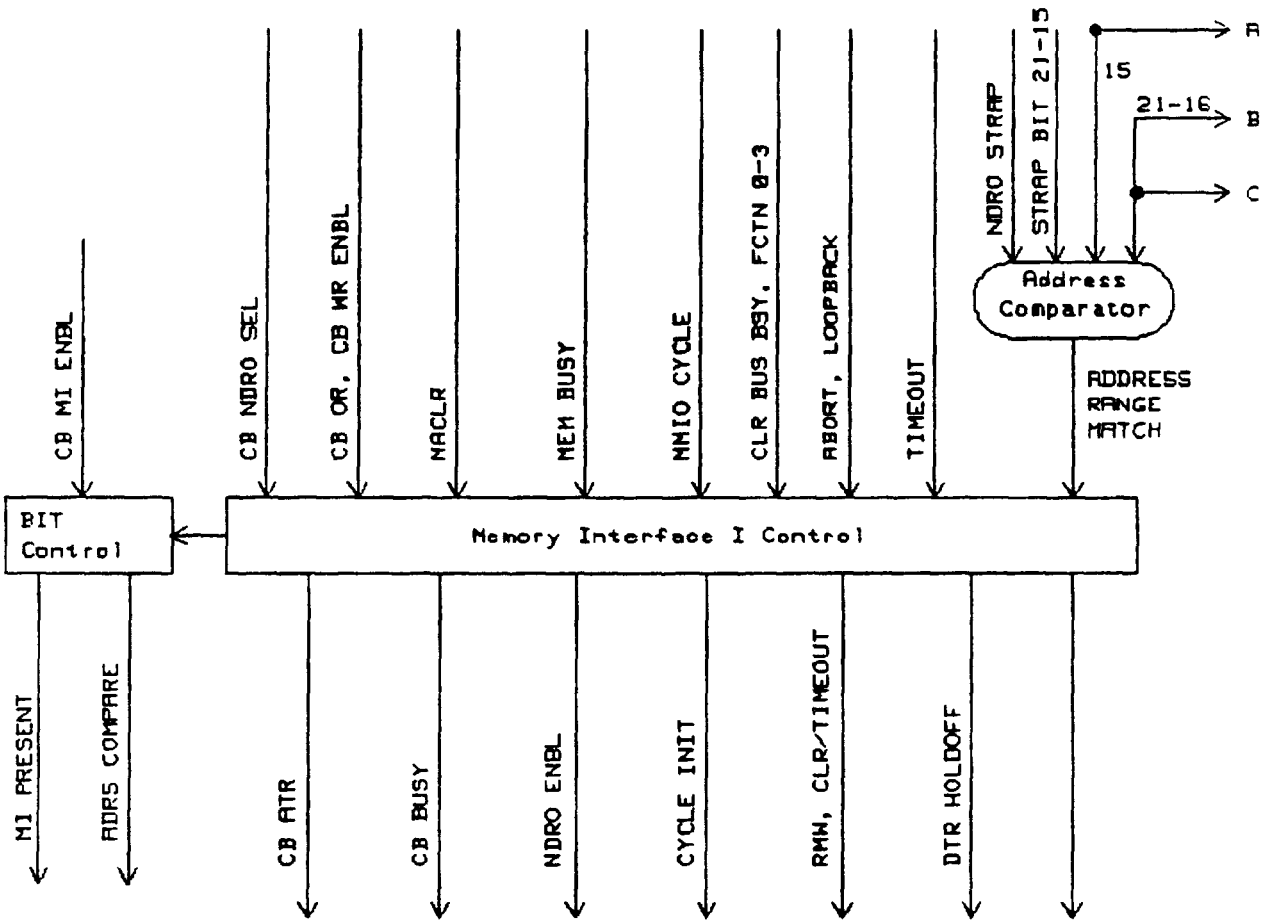


FIGURE 1. CWG functional block diagram.

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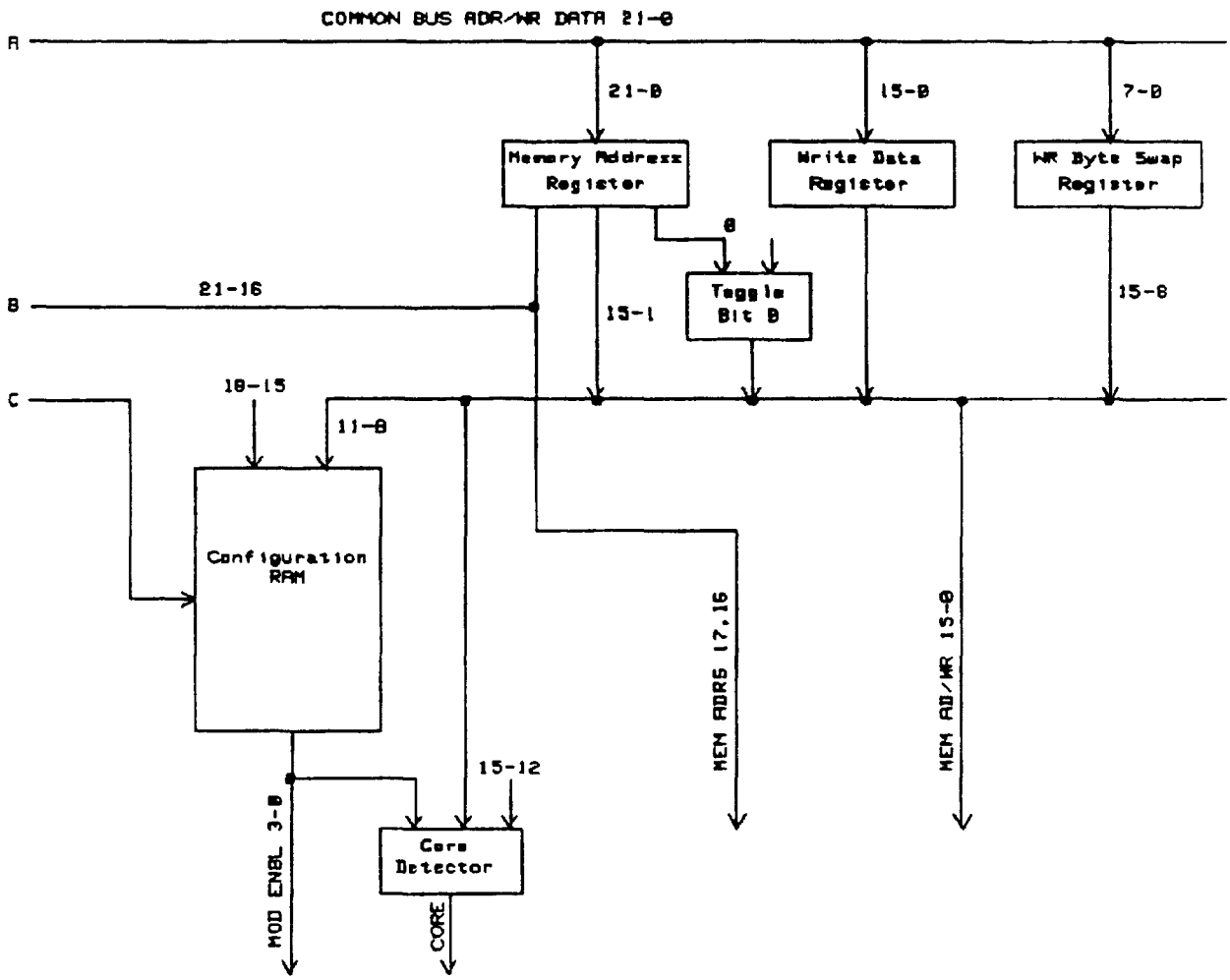


FIGURE 1. CWG functional block diagram - Continued.

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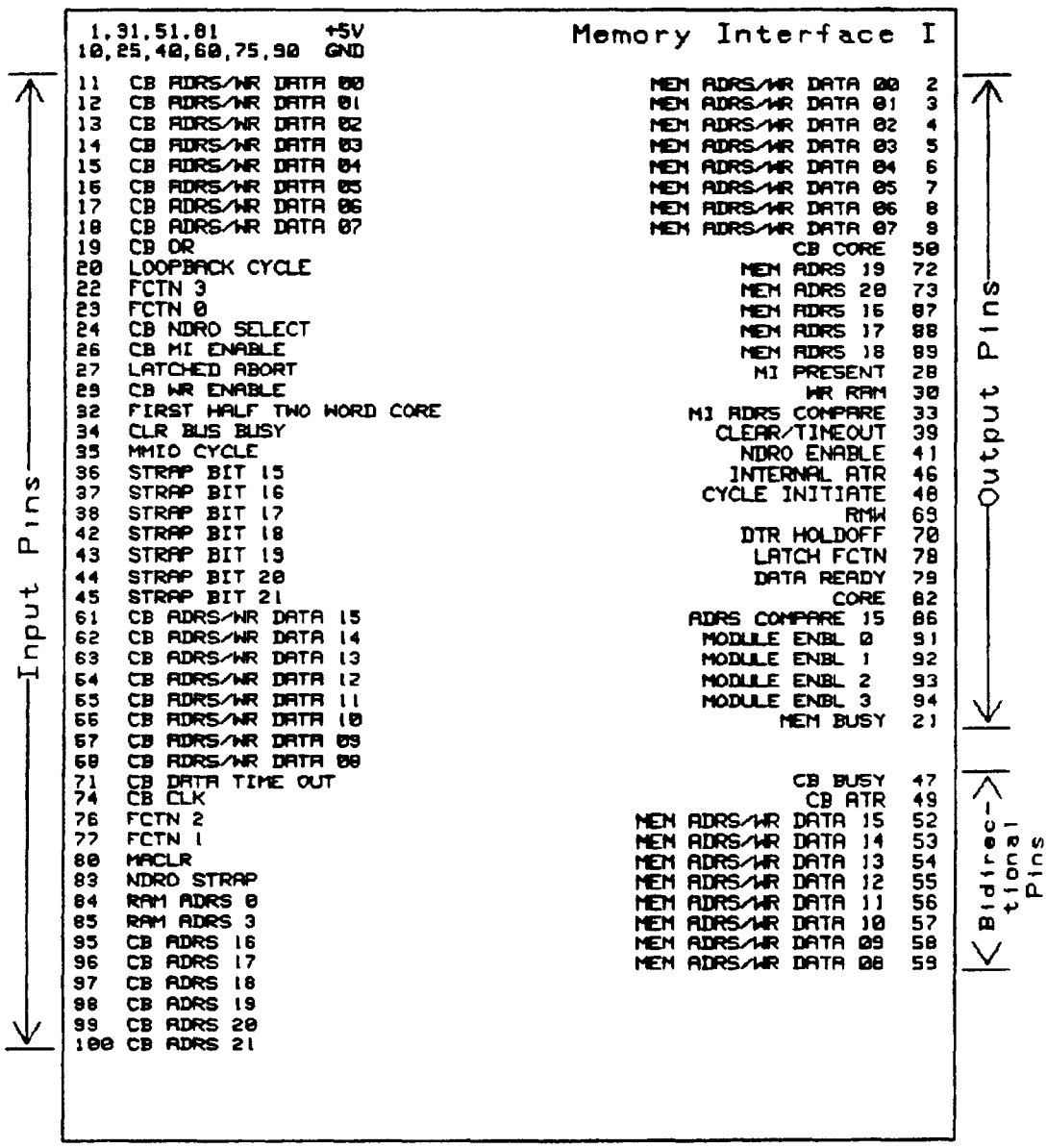


FIGURE 2. CWG pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

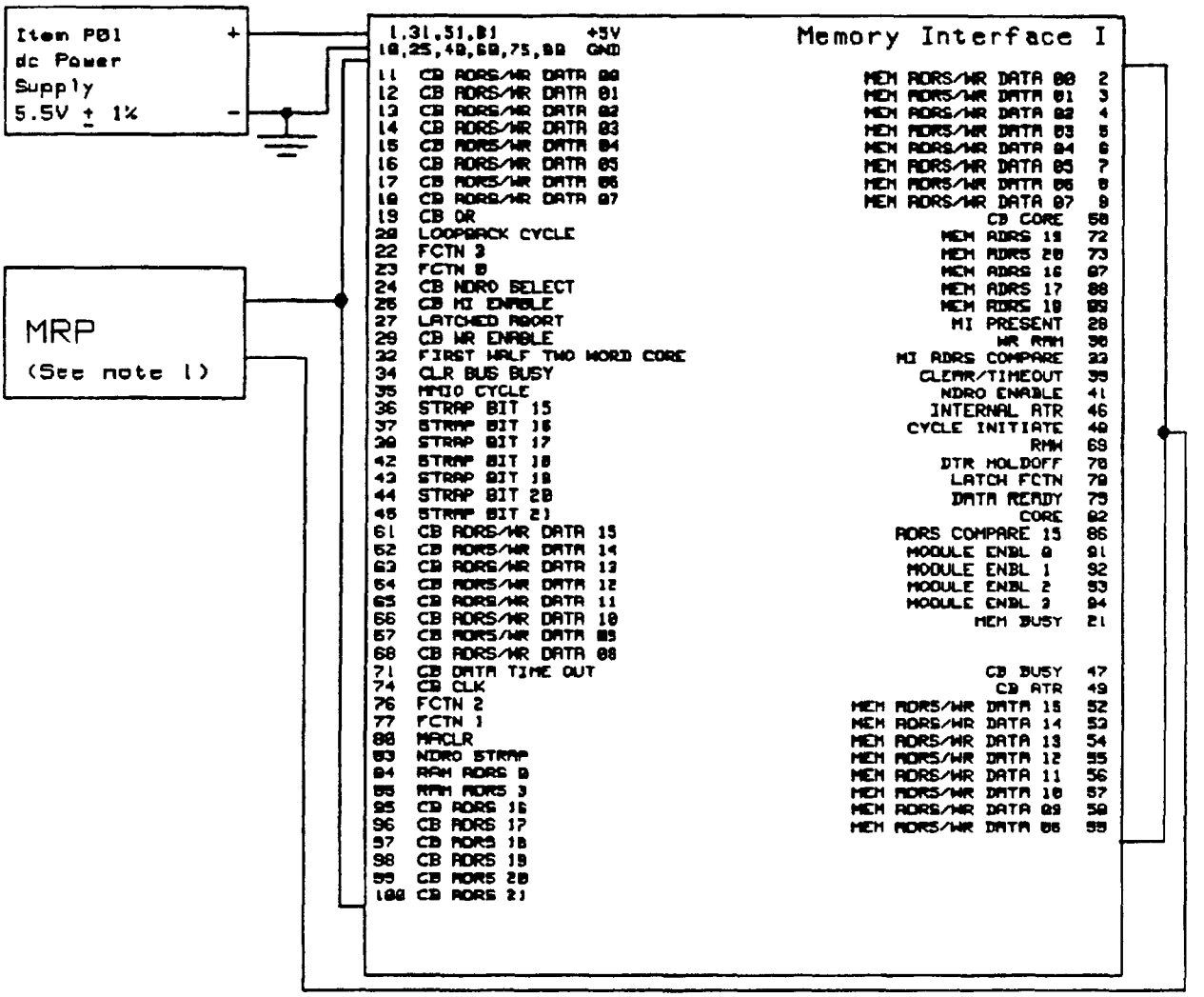
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. CWG operating temperature and component temperature test circuit.

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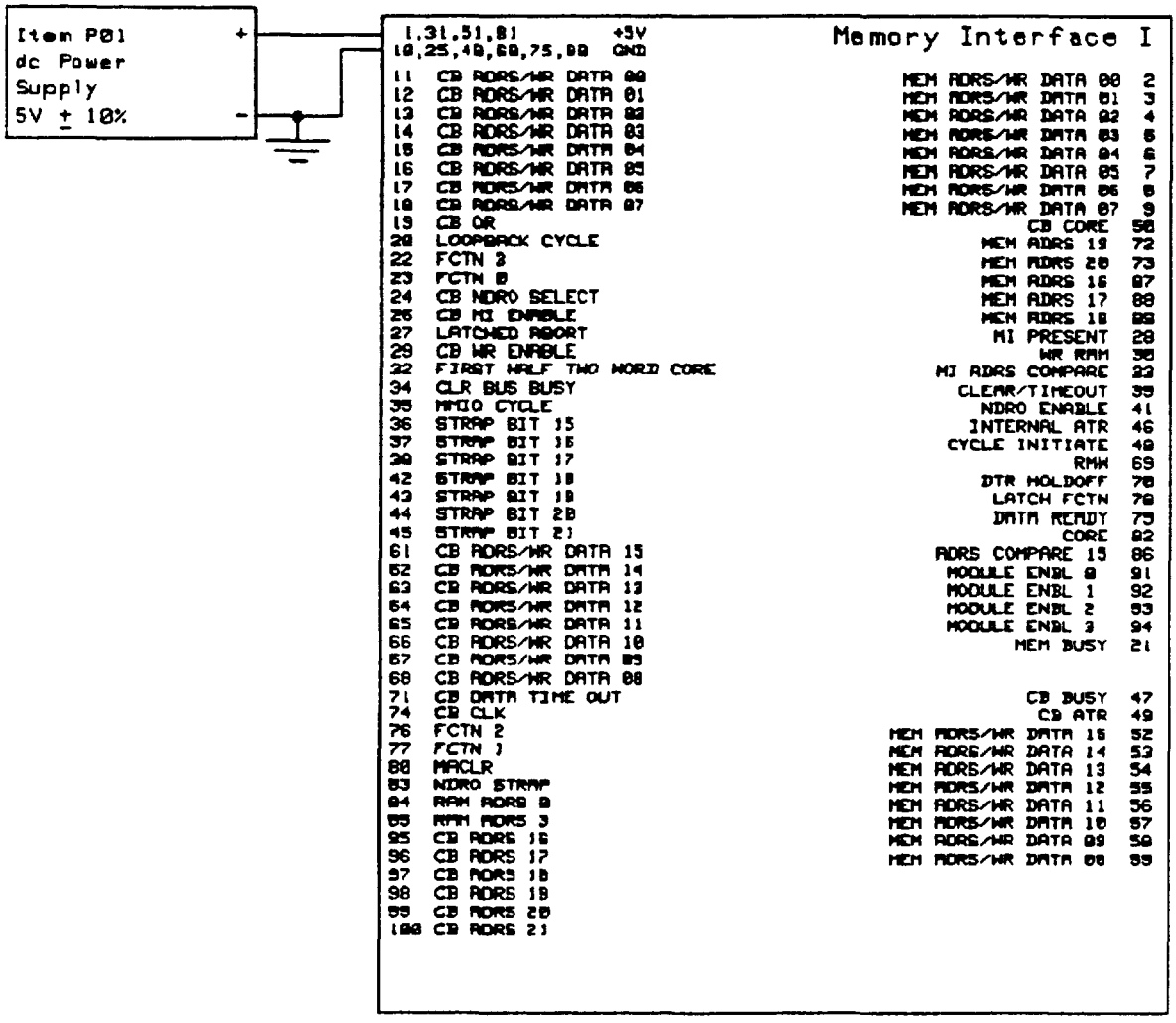


FIGURE 4. CWG vibration test circuit.

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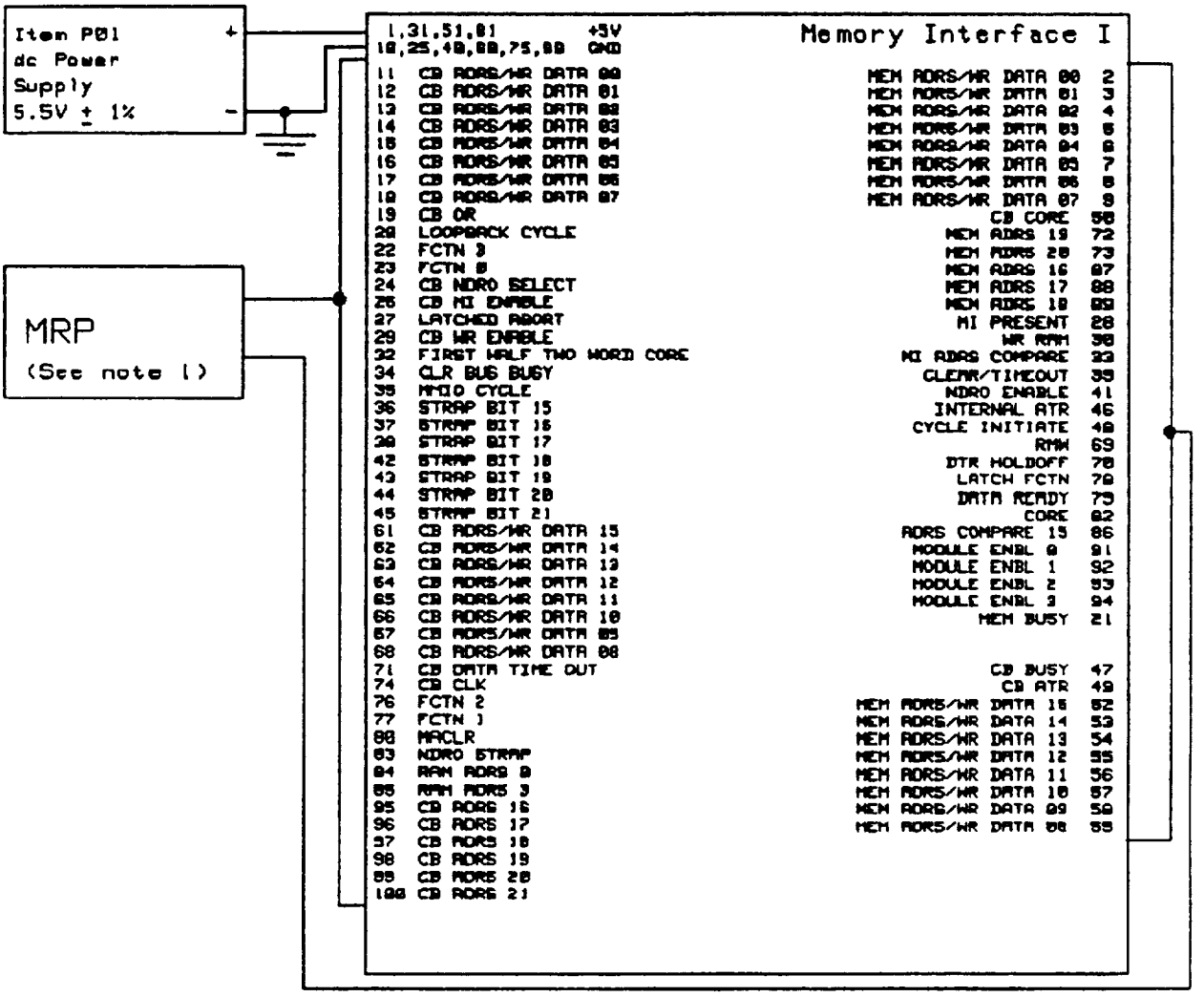
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. CWG life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 6.17 W (max); 4.13 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB AD/WR 00	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 01	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 02	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 03	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 04	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 05	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 06	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 07	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB OR	-1.2	0.4	2.4	-1.2	0.04	0.2	100
LOOPBACK	-1.2	0.4	2.4	-0.6	0.02	0.1	50
FCTN 3	-1.2	0.4	2.4	-4.4	0.13	1.4	125
FCTN 0	-1.2	0.4	2.4	-3.2	0.09	1.2	100
CB NDRO SEL	-1.2	0.4	2.4	-2.4	0.08	0.4	150
CB MI ENBL	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ABORT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB WR ENBL	-1.2	0.4	2.4	-0.6	0.02	0.1	50
FIRST HALF	-1.2	0.4	2.4	-2.4	0.08	0.4	125
CLR BUS BSY	-1.2	0.4	2.4	-1.2	0.04	0.2	75
MMIO CYCLE	NA	0.4	2.4	-8.03	-2.8	1.21	75
STRAP BIT 15	-1.5	0.4	2.4	-0.36	0.02	0.1	50
STRAP BIT 16	-1.5	0.4	2.4	-0.36	0.02	0.1	75
STRAP BIT 17	-1.5	0.4	2.4	-0.36	0.02	0.1	75
STRAP BIT 18	-1.5	0.4	2.4	-0.36	0.02	0.1	50
STRAP BIT 19	-1.5	0.4	2.4	-0.36	0.02	0.1	50
STRAP BIT 20	-1.5	0.4	2.4	-0.36	0.02	0.1	50
STRAP BIT 21	-1.5	0.4	2.4	-0.36	0.02	0.1	50
CB BSY	NA	0.4	2.4	-0.65	0.07	NA	100
CB ATR	NA	0.4	2.4	-0.65	0.07	NA	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB AD/WR 15	-1.2	0.4	2.4	-1.2	0.04	0.2	100
CB AD/WR 14	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 13	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 12	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 11	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 10	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 09	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB AD/WR 08	-1.2	0.4	2.4	-0.6	0.02	0.1	50
TIMEOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	50
FCTN 2	-1.2	0.4	2.4	-3.2	0.09	1.2	100
FCTN 1	-1.2	0.4	2.4	-3.8	0.11	1.3	125
MACLR	-1.2	0.4	2.4	-1.2	0.04	0.2	75
NDRO STRAP	-1.5	0.4	2.4	-0.36	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RAM ADRS 0	-1.2	0.4	2.4	-0.1	0.01	0.1	75
RAM ADRS 3	-1.2	0.4	2.4	-0.1	0.01	0.1	75
CB ADRS 16	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB ADRS 17	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB ADRS 18	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB ADRS 19	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB ADRS 20	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB ADRS 21	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM AD/WR 15	NA	NA	NA	NA	NA	NA	100
MEM AD/WR 14	NA	NA	NA	NA	NA	NA	100
MEM AD/WR 13	NA	NA	NA	NA	NA	NA	125
MEM AD/WR 12	NA	NA	NA	NA	NA	NA	100
MEM AD/WR 11	NA	NA	NA	NA	NA	NA	150

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEM AD/WR 10	NA	NA	NA	NA	NA	NA	150
MEM AD/WR 09	NA	NA	NA	NA	NA	NA	150
MEM AD/WR 08	NA	NA	NA	NA	NA	NA	150

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEM AD/WR 00	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM AD/WR 01	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM AD/WR 02	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM AD/WR 03	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM AD/WR 04	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM AD/WR 05	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM AD/WR 06	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM AD/WR 07	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
CB CORE	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
MEM AD/WR 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
MEM AD/WR 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
MEM AD/WR 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
MEM AD/WR 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
MEM AD/WR 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEM AD/WR 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
MEM AD/WR 09	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
MEM AD/WR 08	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
MEM ADRS 19	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
MEM ADRS 20	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM ADRS 16	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MEM ADRS 17	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
MEM ADRS 18	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
MI PRESENT	NA	0.5	NA	6.0	0.5	NA	NA	NA
WR RAM	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ADRS COMPARE	NA	0.5	NA	6.0	0.25	NA	NA	NA
CLR/TIMEOUT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
NDRO ENBL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
INTERNAL ATR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CYCLE INIT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RMW	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
DTR HOLDOFF	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
LATCH FCTN	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DATA READY	NA	0.5	NA	6.0	0.25	NA	NA	NA
CORE	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
ADRS CMPR 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MOD ENBL 0	2.4	0.5	-0.4	6.0	NA	NA	-12.0	NA
MOD ENBL 1	2.4	0.5	-0.4	6.0	NA	NA	-12.0	NA
MOD ENBL 2	2.4	0.5	-0.4	6.0	NA	NA	-12.0	NA
MOD ENBL 3	2.4	0.5	-0.4	6.0	NA	NA	-12.0	NA
MEM BSY	NA	NA	NA	NA	NA	NA	-28.8	NA
CB BSY	2.4	0.5	-0.4	6.0	NA	NA	-100.0	100
CB ATR	2.4	0.5	-0.4	6.0	NA	NA	-100.0	125

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLK to INTERNAL ATR	-	-	-	23	23	23	nS	12a
2	tpd	CB CLK to CB ATR	-	-	-	33	33	33	nS	12a
3	tpd	CB CLK to ADRS COMPARE	-	-	-	48	48	48	nS	12a
3	tpd	CB CLK to DTR HOLDOFF	-	-	-	48	48	48	nS	12a
4	tpd	CB CLK to MEM AD/WR 00	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 01	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 02	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 03	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 04	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 05	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 06	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 07	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 08	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 09	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 10	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 11	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 12	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 13	-	-	-	38	38	38	nS	12a

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	CB CLK to MEM AD/WR 14	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 15	-	-	-	38	38	38	nS	12a
5	tpd	RAM ADRS 0 to CORE	-	-	-	52	52	52	nS	12a
5	tpd	RAM ADRS 3 to CORE	-	-	-	52	52	52	nS	12a
6	th	CB CLK to CB OR	11	11	11	-	-	-	nS	12a
6	th	CB CLK to TIMEOUT	11	11	11	-	-	-	nS	12a
7	tsu	CB NDRO SEL to CB CLK	12	12	12	-	-	-	nS	12b
8	th	CB CLK to CB NDRO SEL	8	8	8	-	-	-	nS	12b
9	tsu	FCTN 0 to CB CLK	29	29	29	-	-	-	nS	12b
10	th	CB CLK to FCTN 0	26	26	26	-	-	-	nS	12b
11	tsu	ABORT to CB CLK	33	33	33	-	-	-	nS	12b
12	th	CB CLK to ABORT	5	5	5	-	-	-	nS	12b
13	th	CB CLK to CB ADRS 21	65	65	65	-	-	-	nS	12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
14	tsu	LOOPBACK to CB CLK	23	23	23	-	-	-	nS	12b
15	th	CB CLK to LOOPBACK	6	6	6	-	-	-	nS	12b

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6.4 Module function. The memory interface transfers memory addresses and data between the common bus and memory. The memory interface is contained on two modules: memory interface I (CWG) and memory interface II (EWD). Memory interface I controls the address and write data transfers and memory interface II controls read data transfers. Each memory interface provides the interface for a maximum of 1 million words of magnetic core or semiconductor memory to the AN/UYK-44 common bus. Modularity of memory may be 64K (K = 1024), 128K, 256K, or 512K words, with a maximum of four memory modules per memory interface. The maximum UYK-44 memory configuration is four memory interfaces.

6.4.1 Address comparator. The address comparator compares common bus address bits 21-15 with Memory Interface (MI) strap bits 21-15 to determine if the address is within the allocated address range for this memory interface. The strap bits are wired externally to +5 volts, ground, and address bits to define the address range for the memory interface. When a match occurs the address is latched in the memory address register.

6.4.2 Memory address register. The 22-bit register holds the memory address during read and write operations. The address from the common bus is latched when the address comparator detects an address range match. Prior to being latched, memory address register bits 21-16 appear transparent to the address comparator. Memory address register bits 15-0 are placed on the memory address/write data bus to memory. Bit 0 passes through a toggle circuit and is unaltered for all operations except a double word reference to core memory. For this operation, the bit is complemented on the first memory reference and is not altered on the second memory reference. Memory address register bits 18-15 are used to address the configuration Random Access Memory (RAM).

6.4.3 Write data register. The 16-bit register holds memory write data. The register is loaded from the common bus when the data is placed on the bus by the bus master (Central Processor (CP) or Input Output Controller (IOC)), the OR signal is received, and the function code specifies a write operation (F2 = low). The register output is placed on the memory address/write data bus when the memory busy signal is received.

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6.4.4 Write byte swap register. The 8-bit register transfers the lower byte (bits 7-0) from the common bus to the upper byte (bits 15-8) sent to memory. The register is loaded with common bus bits 7-0 when the OR signal is received and the function code specifies a write operation (F2 = low). The register output is placed in memory address/write data bus bits 15-8 when the memory busy signal is received and the function code specifies write byte upper (F2 = low, F1 = high, and F0 = low).

6.4.5 Configuration RAM. The configuration RAM allows the memory interface to interface with a variety of memory module sizes and still maintain addressing continuity. The RAM contains 16 4-bit words. Each bit provides a module enable signal to the corresponding memory module (bit 0 enables module 0, bit 1 enables module 1, and so forth). Only one module enable can be active, therefore, only one bit in each word can be set. The RAM is addressed by the upper address bits. Addressing is variable, however, when address bits 18 through 15 are used, each RAM location is used to address 32K of memory. Therefore, the 16 RAM words can address a maximum of 512K (K = 1024 words) of memory. Memory modules can contain 32K, 64K, or 128K words. The configuration RAM is loaded as part of a power-up routine to reflect the module address range. For example, if module 0 contained 64K words and module 1 contained 32K words, bit 0 in RAM words 0 and 1 would be set to enable module 0 for the 0-64K address range and bit 1 in RAM word 1 would be set to enable module 1 for the 64K to 96K address range. The addressed RAM location is loaded with write data register bits 11-8 by a write configuration RAM function code.

6.4.6 Core detector. The core detector provides a core signal when a memory reference is made to a module containing core memory. The detector consists of a 4-bit register (each bit indicates if the corresponding module contains core) and a compare circuit that compares module enables 3-0 with the corresponding register bits 3-0. The register is loaded with write data register bits 15-12 by a write configuration RAM function code (loaded same time as configuration RAM).

6.4.7 Built-In-Test (BIT) control. BIT control provides the maintenance processor (BWAH) with information. The maintenance processor activates BIT control with the CB MI enable signal. BIT control responds with the MI present signal (low). It also responds with the MI address compare signal (low) if the memory interface has been addressed via the common bus.

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6.4.8 Memory interface I and II control. The MI I and II control circuits provide the control to transfer addresses and data between the common bus and memory modules.

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6.4.9 Description of I/O pins.

Signal name	Pin number	I/O	Description
CB AD/WR 00	11	I	CB ADRS/WR DATA 00
CB AD/WR 01	12	I	CB ADRS/WR DATA 01
CB AD/WR 02	13	I	CB ADRS/WR DATA 02
CB AD/WR 03	14	I	CB ADRS/WR DATA 03
CB AD/WR 04	15	I	CB ADRS/WR DATA 04
CB AD/WR 05	16	I	CB ADRS/WR DATA 05
CB AD/WR 06	17	I	CB ADRS/WR DATA 06
CB AD/WR 07	18	I	CB ADRS/WR DATA 07
CB OR	19	I	CB OR
LOOPBACK	20	I	LOOPBACK CYCLE
FCTN 3	22	I	FCTN 3
FCTN 0	23	I	FCTN 0
CB NDRO SEL	24	I	CB NDRO SELECT
CB MI ENBL	26	I	CB MI ENABLE
ABORT	27	I	LATCHED ABORT
CB WR ENBL	29	I	CB WR ENABLE
FIRST HALF	32	I	FIRST HALF TWO WORD CORE
CLR BUS BSY	34	I	CLR BUS BUSY
MMIO CYCLE	35	I	MMIO CYCLE
STRAP BIT 15	36	I	STRAP BIT 15

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6.4.9 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
STRAP BIT 16	37	I	STRAP BIT 16
STRAP BIT 17	38	I	STRAP BIT 17
STRAP BIT 18	42	I	STRAP BIT 18
STRAP BIT 19	43	I	STRAP BIT 19
STRAP BIT 20	44	I	STRAP BIT 20
STRAP BIT 21	45	I	STRAP BIT 21
CB BSY	47	I/O	CB BUSY
CB ATR	49	I/O	CB ATR
CB AD/WR 15	61	I	CB ADRS/WR DATA 15
CB AD/WR 14	62	I	CB ADRS/WR DATA 14
CB AD/WR 13	63	I	CB ADRS/WR DATA 13
CB AD/WR 12	64	I	CB ADRS/WR DATA 12
CB AD/WR 11	65	I	CB ADRS/WR DATA 11
CB AD/WR 10	66	I	CB ADRS/WR DATA 10
CB AD/WR 09	67	I	CB ADRS/WR DATA 09
CB AD/WR 08	68	I	CB ADRS/WR DATA 08
TIMEOUT	71	I	CB DATA TIME OUT
CB CLK	74	I	CB CLK
FCTN 2	76	I	FCTN 2
FCTN 1	77	I	FCTN 1

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6.4.9 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MACLR	80	I	MACLR
NDRO STRAP	83	I	NDRO STRAP
RAM ADRS 0	84	I	RAM ADRS 0
RAM ADRS 3	85	I	RAM ADRS 3
CB ADRS 16	95	I	CB ADRS 16
CB ADRS 17	96	I	CB ADRS 17
CB ADRS 18	97	I	CB ADRS 18
CB ADRS 19	98	I	CB ADRS 19
CB ADRS 20	99	I	CB ADRS 20
CB ADRS 21	100	I	CB ADRS 21
MEM AD/WR 00	2	O	MEM ADRS/WR DATA 00
MEM AD/WR 01	3	O	MEM ADRS/WR DATA 01
MEM AD/WR 02	4	O	MEM ADRS/WR DATA 02
MEM AD/WR 03	5	O	MEM ADRS/WR DATA 03
MEM AD/WR 04	6	O	MEM ADRS/WR DATA 04
MEM AD/WR 05	7	O	MEM ADRS/WR DATA 05
MEM AD/WR 06	8	O	MEM ADRS/WR DATA 06
MEM AD/WR 07	9	O	MEM ADRS/WR DATA 07
CB CORE	50	O	CB CORE
MEM AD/WR 15	52	I/O	MEM ADRS/WR DATA 15

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6.4.9 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEM AD/WR 14	53	I/O	MEM ADRS/WR DATA 14
MEM AD/WR 13	54	I/O	MEM ADRS/WR DATA 13
MEM AD/WR 12	55	I/O	MEM ADRS/WR DATA 12
MEM AD/WR 11	56	I/O	MEM ADRS/WR DATA 11
MEM AD/WR 10	57	I/O	MEM ADRS/WR DATA 10
MEM AD/WR 09	58	I/O	MEM ADRS/WR DATA 09
MEM AD/WR 08	59	I/O	MEM ADRS/WR DATA 08
MEM ADRS 19	72	O	MEM ADRS 19
MEM ADRS 20	73	O	MEM ADRS 20
MEM ADRS 16	87	O	MEM ADRS 16
MEM ADRS 17	88	O	MEM ADRS 17
MEM ADRS 18	89	O	MEM ADRS 18
MI PRESENT	28	O	MI PRESENT
WR RAM	30	O	WR RAM
ADRS COMPARE	33	O	MI ADRS COMPARE
CLR/TIMEOUT	39	O	CLEAR/TIMEOUT
NDRO ENBL	41	O	NDRO ENABLE
INTERNAL ATR	46	O	INTERNAL ATR
CYCLE INIT	48	O	CYCLE INITIATE
RMW	69	O	RMW

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6.4.9 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
DTR HOLDOFF	70	O	DTR HOLDOFF
LATCH FCTN	78	O	LATCH FCTN
DATA READY	79	O	DATA READY
CORE	82	O	CORE
ADRS CMPR 15	86	O	ADRS COMPARE 15
MOD ENBL 0	91	O	MODULE ENBL 0
MOD ENBL 1	92	O	MODULE ENBL 1
MOD ENBL 2	93	O	MODULE ENBL 2
MOD ENBL 3	94	O	MODULE ENBL 3
MEM BSY	21	O	MEM BUSY

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N735)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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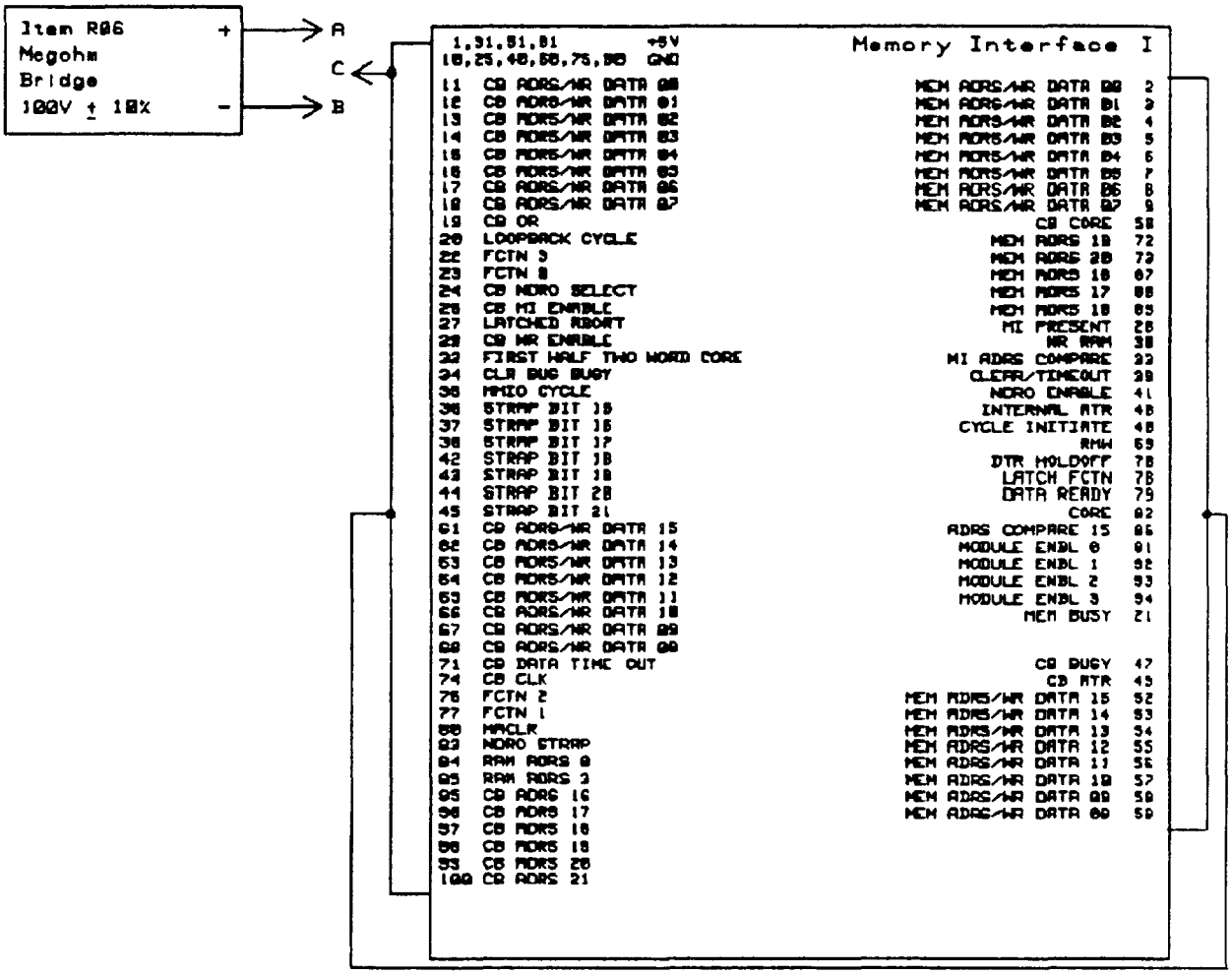


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

#063+

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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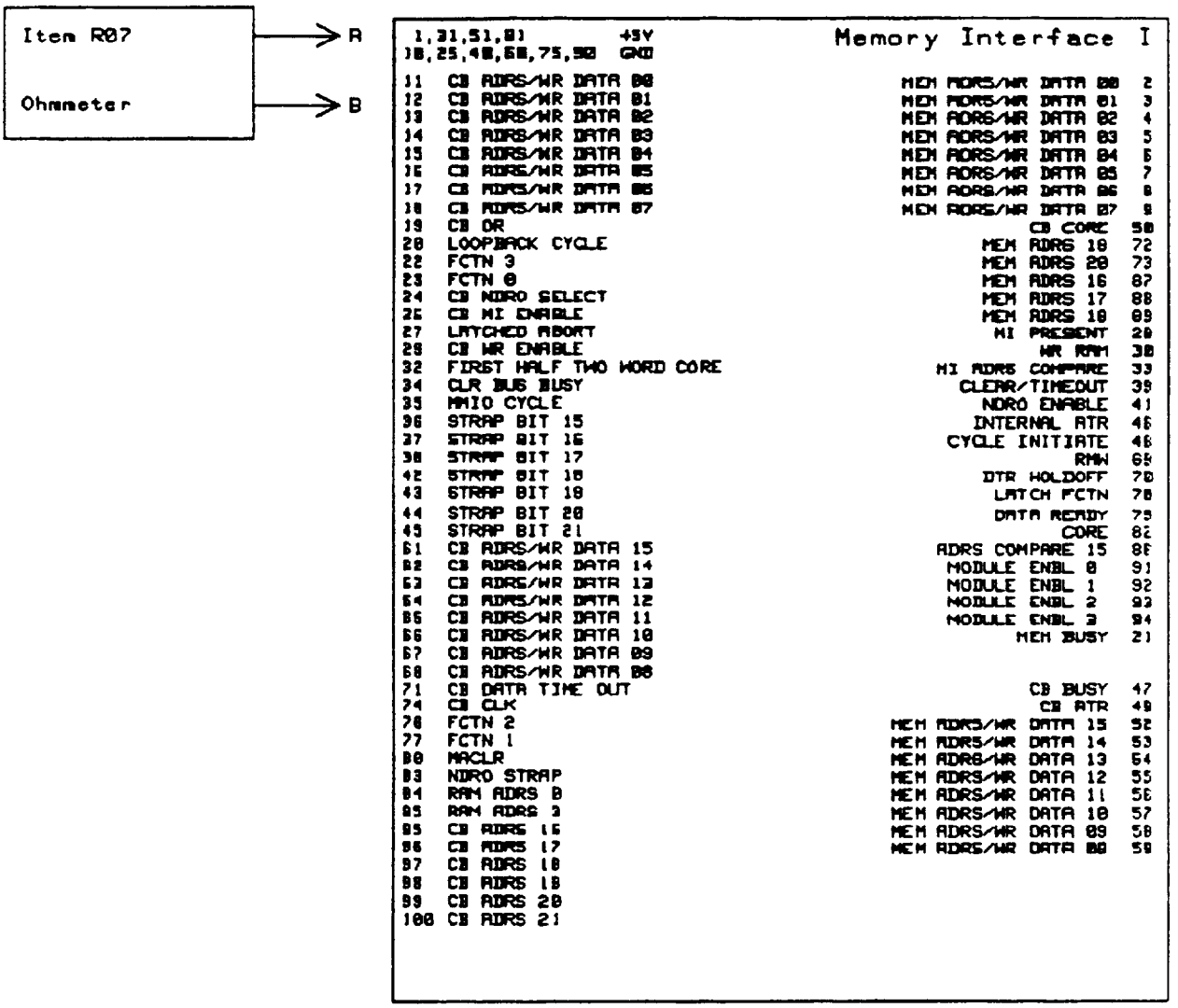


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all re-maining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.02 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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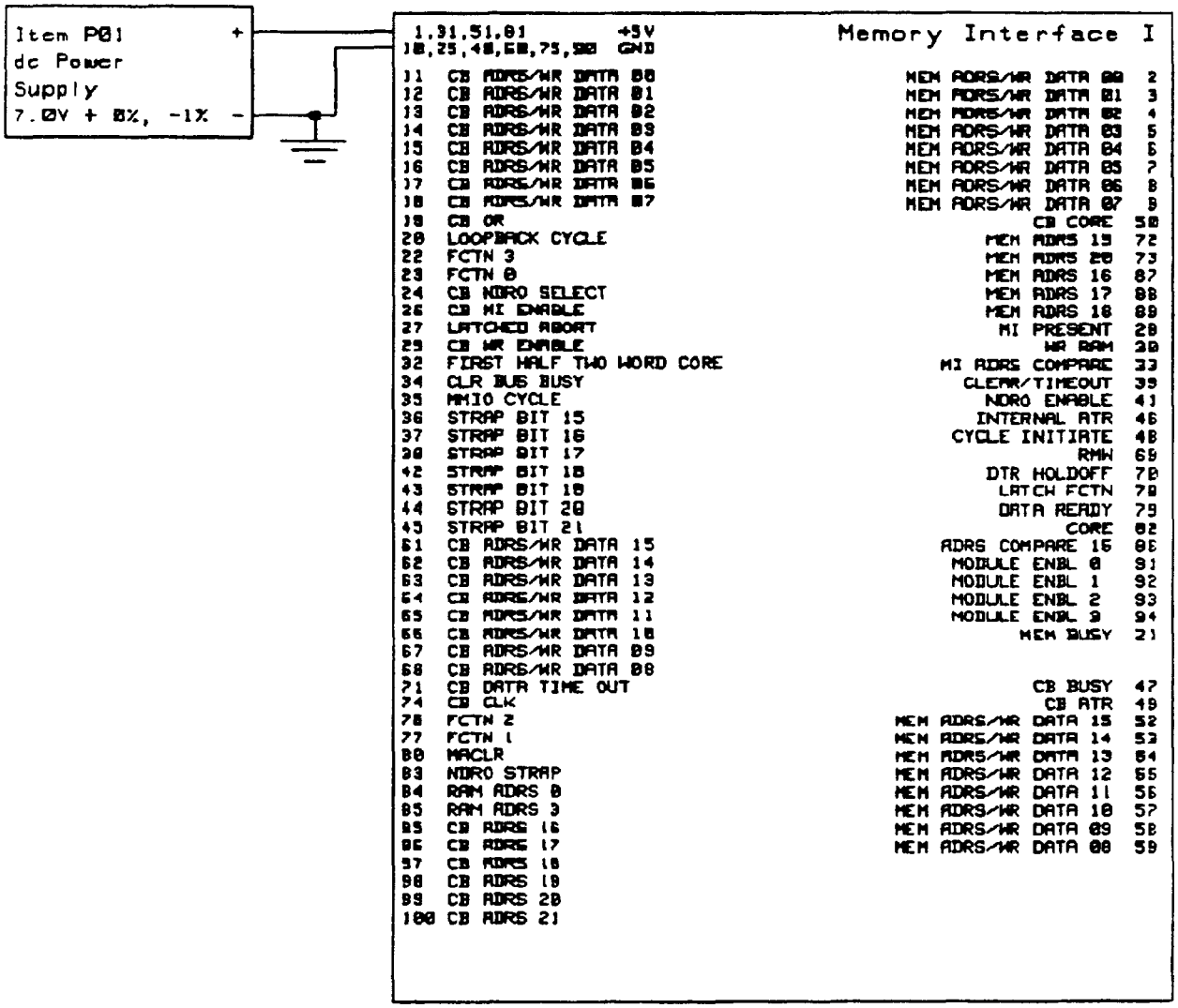


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	CB AD/WR 00	11	5.5	0.0	0.1	mA	1-3
D2	CB AD/WR 01	12					
D3	CB AD/WR 02	13					
D4	CB AD/WR 03	14					
D5	CB AD/WR 04	15					
D6	CB AD/WR 05	16					
D7	CB AD/WR 06	17					
D8	CB AD/WR 07	18	5.5	0.0	0.1	mA	1-3
D9	CB OR	19	5.5	0.0	0.2	mA	1-3
D10	LOOPBACK	20	5.5	0.0	0.1	mA	1-3
D11	FCTN 3	22	5.5	0.0	1.4	mA	1-3
D12	FCTN 0	23	5.5	0.0	1.2	mA	1-3
D13	CB NDRO SEL	24	5.5	0.0	0.4	mA	1-3
D14	CB MI ENBL	26	5.5	0.0	0.1	mA	1-3
D15	ABORT	27					
D16	CB WR ENBL	29	5.5	0.0	0.1	mA	1-3
D17	FIRST HALF	32	5.5	0.0	0.4	mA	1-3
D18	CLR BUS BSY	34	5.5	0.0	0.2	mA	1-3
D19	MMIO CYCLE	35	5.5	-0.12	1.21	mA	1-3
D20	STRAP BIT 15	36	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	STRAP BIT 16	37	5.5	0.0	0.1	mA	1-3
D22	STRAP BIT 17	38					
D23	STRAP BIT 18	42					
D24	STRAP BIT 19	43					
D25	STRAP BIT 20	44					
D26	STRAP BIT 21	45	5.5	0.0	0.1	mA	1-3
D27	CB AD/WR 15	61	5.5	0.0	0.2	mA	1-3
D28	CB AD/WR 14	62	5.5	0.0	0.1	mA	1-3
D29	CB AD/WR 13	63					
D30	CB AD/WR 12	64					
D31	CB AD/WR 11	65					
D32	CB AD/WR 10	66					
D33	CB AD/WR 09	67					
D34	CB AD/WR 08	68					
D35	TIMEOUT	71	5.5	0.0	0.1	mA	1-3
D36	CB CLK	74	5.5	0.0	1.0	mA	1-3
D37	FCTN 2	76	5.5	0.0	1.2	mA	1-3
D38	FCTN 1	77	5.5	0.0	1.3	mA	1-3
D39	MACLR	80	5.5	0.0	0.2	mA	1-3
D40	NDRO STRAP	83	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D41	RAM ADRS 0	84	5.5	0.0	0.1	mA	1-3
D42	RAM ADRS 3	85					
D43	CB ADRS 16	95					
D44	CB ADRS 17	96					
D45	CB ADRS 18	97					
D46	CB ADRS 19	98					
D47	CB ADRS 20	99					
D48	CB ADRS 21	100	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D48
Power supply current limit	1.68 A	NA	D1-D48
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D48
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D48 D1-D48
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 5.0 uA ± 10.0 uA ± 5.0 uA ± 70.0 uA ± 60.0 uA ± 20.0 uA ± 5.0 uA ± 20.0 uA ± 10.0 uA ± 66.5 uA ± 5.0 uA ± 10.0 uA ± 5.0 uA ± 50.0 uA ± 60.0 uA ± 65.0 uA ± 10.0 uA ± 5.0 uA	D1-D8 D9 D10 D11 D12 D13 D14-D16 D17 D18 D19 D20-D26 D27 D28-D35 D36 D37 D38 D39 D40-D48

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	MEM AD/WR 00	2	0.0	-150.0	-60.0	mA	1-4
E2	MEM AD/WR 00	2					
E3	MEM AD/WR 01	3					
E4	MEM AD/WR 01	3					
E5	MEM AD/WR 02	4					
E6	MEM AD/WR 02	4					
E7	MEM AD/WR 03	5					
E8	MEM AD/WR 03	5					
E9	MEM AD/WR 04	6					
E10	MEM AD/WR 04	6					
E11	MEM AD/WR 05	7					
E12	MEM AD/WR 05	7					
E13	MEM AD/WR 06	8					
E14	MEM AD/WR 06	8					
E15	MEM AD/WR 07	9					
E16	MEM AD/WR 07	9					
E17	WR RAM	30					
E18	CLR/TIMEOUT	39					
E19	NDRO ENBL	41					
E20	INTERNAL ATR	46	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	CB BSY	47	0.0	-325.0	-100.0	mA	1-4
E22	CYCLE INIT	48	0.0	-150.0	-60.0	mA	1-4
E23	CB ATR	49	0.0	-325.0	-100.0	mA	1-4
E24	CB CORE	50	0.0	-325.0	-100.0	mA	1-4
E25	MEM AD/WR 15	52	0.0	-150.0	-60.0	mA	1-4
E26	MEM AD/WR 15	52					
E27	MEM AD/WR 15	52					
E28	MEM AD/WR 14	53					
E29	MEM AD/WR 14	53					
E30	MEM AD/WR 14	53					
E31	MEM AD/WR 13	54					
E32	MEM AD/WR 13	54					
E33	MEM AD/WR 13	54					
E34	MEM AD/WR 12	55					
E35	MEM AD/WR 12	55					
E36	MEM AD/WR 12	55					
E37	MEM AD/WR 11	56					
E38	MEM AD/WR 11	56					
E39	MEM AD/WR 11	56					
E40	MEM AD/WR 10	57	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	MEM AD/WR 10	57	0.0	-150.0	-60.0	mA	1-4
E42	MEM AD/WR 10	57					
E43	MEM AD/WR 09	58					
E44	MEM AD/WR 09	58					
E45	MEM AD/WR 09	58					
E46	MEM AD/WR 08	59					
E47	MEM AD/WR 08	59					
E48	MEM AD/WR 08	59	0.0	-150.0	-60.0	mA	1-4
E49	RMW	69	0.0	-100.0	-40.0	mA	1-4
E50	DTR HOLDOFF	70	0.0	-100.0	-40.0	mA	1-4
E51	MEM ADRS 19	72	0.0	-150.0	-60.0	mA	1-4
E52	MEM ADRS 20	73					
E53	LATCH FCTN	78	0.0	-150.0	-60.0	mA	1-4
E54	CORE	82	0.0	-100.0	-40.0	mA	1-4
E55	ADRS CMPR 15	86	0.0	-150.0	-60.0	mA	1-4
E56	MEM ADRS 16	87					
E57	MEM ADRS 17	88					
E58	MEM ADRS 18	89	0.0	-150.0	-60.0	mA	1-4
E59	MOD ENBL 0	91	0.0	-90.0	-12.0	mA	1-4
E60	MOD ENBL 1	92	0.0	-90.0	-12.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E61	MOD ENBL 2	93	0.0	-90.0	-12.0	mA	1-4
E62	MOD ENBL 3	94	0.0	-90.0	-12.0	mA	1-4
E63	MEM BSY	21	0.0	-68.5	-28.8	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E63
Power supply current limit	1.68 A	NA	E1-E63
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E63
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E63 E1-E63
Output current measurement	NA NA NA NA NA NA NA NA NA NA NA	± 4.5 mA ± 11.2 mA ± 4.5 mA ± 11.2 mA ± 4.5 mA ± 3.0 mA ± 4.5 mA ± 3.0 mA ± 4.5 mA ± 3.9 mA ± 1.98 mA	E1-E20 E21 E22 E23,E24 E25-E48 E49,E50 E51-E53 E54 E55-E58 E59-E62 E63

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	CB AD/WR 00	11	2.4	-0.6	0.02	mA	1-3
F2	CB AD/WR 01	12					
F3	CB AD/WR 02	13					
F4	CB AD/WR 03	14					
F5	CB AD/WR 04	15					
F6	CB AD/WR 05	16					
F7	CB AD/WR 06	17					
F8	CB AD/WR 07	18	2.4	-0.6	0.02	mA	1-3
F9	CB OR	19	2.4	-1.2	0.04	mA	1-3
F10	LOOPBACK	20	2.4	-0.6	0.02	mA	1-3
F11	FCTN 3	22	2.4	-4.4	0.13	mA	1-3
F12	FCTN 0	23	2.4	-3.2	0.09	mA	1-3
F13	CB NDRO SEL	24	2.4	-2.4	0.08	mA	1-3
F14	CB MI ENBL	26	2.4	-0.6	0.02	mA	1-3
F15	ABORT	27					
F16	CB WR ENBL	29	2.4	-0.6	0.02	mA	1-3
F17	FIRST HALF	32	2.4	-2.4	0.08	mA	1-3
F18	CLR BUS BSY	34	2.4	-1.2	0.04	mA	1-3
F19	MMIO CYCLE	35	2.4	-8.03	-2.8	mA	1-3
F20	STRAP BIT 15	36	2.4	-0.36	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	STRAP BIT 16	37	2.4	-0.36	0.02	mA	1-3
F22	STRAP BIT 17	38					
F23	STRAP BIT 18	42					
F24	STRAP BIT 19	43					
F25	STRAP BIT 20	44					
F26	STRAP BIT 21	45	2.4	-0.36	0.02	mA	1-3
F27	CB BSY	47	2.4	-0.65	0.07	mA	1-3
F28	CB ATR	49	2.4	-0.65	0.07	mA	1-3
F29	CB AD/WR 15	61	2.4	-1.2	0.04	mA	1-3
F30	CB AD/WR 14	62	2.4	-0.6	0.02	mA	1-3
F31	CB AD/WR 13	63					
F32	CB AD/WR 12	64					
F33	CB AD/WR 11	65					
F34	CB AD/WR 10	66					
F35	CB AD/WR 09	67					
F36	CB AD/WR 08	68					
F37	TIMEOUT	71	2.4	-0.6	0.02	mA	1-3
F38	CB CLK	74	2.4	-2.0	0.05	mA	1-3
F39	FCTN 2	76	2.4	-3.2	0.09	mA	1-3
F40	FCTN 1	77	2.4	-3.8	0.11	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	MACLR	80	2.4	-1.2	0.04	mA	1-3
F42	NDRO STRAP	83	2.4	-0.36	0.02	mA	1-3
F43	RAM ADRS 0	84	2.4	-0.1	0.01	mA	1-3
F44	RAM ADRS 3	85	2.4	-0.1	0.01	mA	1-3
F45	CB ADRS 16	95	2.4	-0.6	0.02	mA	1-3
F46	CB ADRS 17	96					
F47	CB ADRS 18	97					
F48	CB ADRS 19	98					
F49	CB ADRS 20	99					
F50	CB ADRS 21	100	2.4	-0.6	0.02	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F50
Power supply current limit	1.68 A	NA	F1-F50
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F50
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F50 F1-F50
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 31.0 uA ± 62.0 uA ± 31.0 uA ± 226.0 uA ± 164.0 uA ± 124.0 uA ± 31.0 uA ± 124.0 uA ± 62.0 uA ± 261.0 uA ± 19.0 uA ± 36.0 uA ± 62.0 uA ± 31.0 uA ± 102.0 uA ± 164.0 uA ± 195.0 uA ± 62.0 uA ± 19.0 uA ± 5.5 uA ± 31.0 uA	F1-F8 F9 F10 F11 F12 F13 F14-F16 F17 F18 F19 F20-F26 F27,F28 F29 F30-F37 F38 F39 F40 F41 F42 F43,F44 F45-F50

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	CB AD/WR 00	11	0.4	-0.6	0.02	mA	1-3
G2	CB AD/WR 01	12					
G3	CB AD/WR 02	13					
G4	CB AD/WR 03	14					
G5	CB AD/WR 04	15					
G6	CB AD/WR 05	16					
G7	CB AD/WR 06	17					
G8	CB AD/WR 07	18	0.4	-0.6	0.02	mA	1-3
G9	CB OR	19	0.4	-1.2	0.04	mA	1-3
G10	LOOPBACK	20	0.4	-0.6	0.02	mA	1-3
G11	FCTN 3	22	0.4	-4.4	0.13	mA	1-3
G12	FCTN 0	23	0.4	-3.2	0.09	mA	1-3
G13	CB NDRO SEL	24	0.4	-2.4	0.08	mA	1-3
G14	CB MI ENBL	26	0.4	-0.6	0.02	mA	1-3
G15	ABORT	27					
G16	CB WR ENBL	29	0.4	-0.6	0.02	mA	1-3
G17	FIRST HALF	32	0.4	-2.4	0.08	mA	1-3
G18	CLR BUS BSY	34	0.4	-1.2	0.04	mA	1-3
G19	MMIO CYCLE	35	0.4	-8.03	-2.8	mA	1-3
G20	STRAP BIT 15	36	0.4	-0.36	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	STRAP BIT 16	37	0.4	-0.36	0.02	mA	1-3
G22	STRAP BIT 17	38					
G23	STRAP BIT 18	42					
G24	STRAP BIT 19	43					
G25	STRAP BIT 20	44					
G26	STRAP BIT 21	45	0.4	-0.36	0.02	mA	1-3
G27	CB BSY	47	0.4	-0.65	0.07	mA	1-3
G28	CB ATR	49	0.4	-0.65	0.07	mA	1-3
G29	CB AD/WR 15	61	0.4	-1.2	0.04	mA	1-3
G30	CB AD/WR 14	62	0.4	-0.6	0.02	mA	1-3
G31	CB AD/WR 13	63					
G32	CB AD/WR 12	64					
G33	CB AD/WR 11	65					
G34	CB AD/WR 10	66					
G35	CB AD/WR 09	67					
G36	CB AD/WR 08	68					
G37	TIMEOUT	71	0.4	-0.6	0.02	mA	1-3
G38	CB CLK	74	0.4	-2.0	0.05	mA	1-3
G39	FCTN 2	76	0.4	-3.2	0.09	mA	1-3
G40	FCTN 1	77	0.4	-3.8	0.11	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	MACLR	80	0.4	-1.2	0.04	mA	1-3
G42	NDRO STRAP	83	0.4	-0.36	0.02	mA	1-3
G43	RAM ADRS 0	84	0.4	-0.1	0.01	mA	1-3
G44	RAM ADRS 3	85	0.4	-0.1	0.01	mA	1-3
G45	CB ADRS 16	95	0.4	-0.6	0.02	mA	1-3
G46	CB ADRS 17	96					
G47	CB ADRS 18	97					
G48	CB ADRS 19	98					
G49	CB ADRS 20	99					
G50	CB ADRS 21	100	0.4	-0.6	0.02	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G50
Power supply current limit	1.68 A	NA	G1-G50
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G50
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G50 G1-G50
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 31.0 uA ± 62.0 uA ± 31.0 uA ± 226.0 uA ± 164.0 uA ± 124.0 uA ± 31.0 uA ± 124.0 uA ± 62.0 uA ± 261.0 uA ± 19.0 uA ± 36.0 uA ± 62.0 uA ± 31.0 uA ± 102.0 uA ± 164.0 uA ± 195.0 uA ± 62.0 uA ± 19.0 uA ± 5.5 uA ± 31.0 uA	G1-G8 G9 G10 G11 G12 G13 G14-G16 G17 G18 G19 G20-G26 G27,G28 G29 G30-G37 G38 G39 G40 G41 G42 G43,G44 G45-G50

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	MEM AD/WR 00	2	-0.4	2.4	4.5	V	1-4
H2	MEM AD/WR 00	2					
H3	MEM AD/WR 01	3					
H4	MEM AD/WR 01	3					
H5	MEM AD/WR 02	4					
H6	MEM AD/WR 02	4					
H7	MEM AD/WR 03	5					
H8	MEM AD/WR 03	5					
H9	MEM AD/WR 04	6					
H10	MEM AD/WR 04	6					
H11	MEM AD/WR 05	7					
H12	MEM AD/WR 05	7					
H13	MEM AD/WR 06	8					
H14	MEM AD/WR 06	8					
H15	MEM AD/WR 07	9					
H16	MEM AD/WR 07	9					
H17	WR RAM	30					
H18	CLR/TIMEOUT	39					
H19	NDRO ENBL	41					
H20	INTERNAL ATR	46	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H21	CB BSY	47	-0.4	2.4	4.5	V	1-4
H22	CYCLE INIT	48					
H23	CB ATR	49					
H24	CB CORE	50					
H25	MEM AD/WR 15	52					
H26	MEM AD/WR 15	52					
H27	MEM AD/WR 15	52					
H28	MEM AD/WR 14	53					
H29	MEM AD/WR 14	53					
H30	MEM AD/WR 14	53					
H31	MEM AD/WR 13	54					
H32	MEM AD/WR 13	54					
H33	MEM AD/WR 13	54					
H34	MEM AD/WR 12	55					
H35	MEM AD/WR 12	55					
H36	MEM AD/WR 12	55					
H37	MEM AD/WR 11	56					
H38	MEM AD/WR 11	56					
H39	MEM AD/WR 11	56					
H40	MEM AD/WR 10	57	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	MEM AD/WR 10	57	-0.4	2.4	4.5	V	1-4
H42	MEM AD/WR 10	57					
H43	MEM AD/WR 09	58					
H44	MEM AD/WR 09	58					
H45	MEM AD/WR 09	58					
H46	MEM AD/WR 08	59					
H47	MEM AD/WR 08	59					
H48	MEM AD/WR 08	59					
H49	RMW	69					
H50	DTR HOLDOFF	70					
H51	MEM ADRS 19	72					
H52	MEM ADRS 20	73					
H53	LATCH FCTN	78					
H54	CORE	82					
H55	ADRS CMPR 15	86					
H56	MEM ADRS 16	87					
H57	MEM ADRS 17	88					
H58	MEM ADRS 18	89					
H59	MOD ENBL 0	91					
H60	MOD ENBL 1	92	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H61	MOD ENBL 2	93	-0.4	2.4	4.5	V	1-4
H62	MOD ENBL 3	94	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H62
Power supply current limit	1.68 A	NA	H1-H62
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H62
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H62 H1-H62
Output voltage measurement	NA	± 105 mV	H1-H62

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

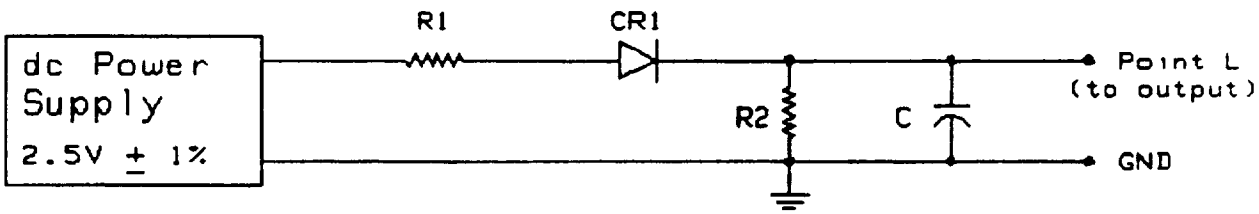
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,9, 30,39,41,46,47, 48,49,50,52,53, 54,55,56,57,58, 59,69,70,72,73, 78,82,86,87,88, 89,91,92,93,94	6200	6.0	0.5
28,33,79	NA	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	MEM AD/WR 00	2	6.0	0.0	0.5	V	1-4
J2	MEM AD/WR 00	2					
J3	MEM AD/WR 01	3					
J4	MEM AD/WR 01	3					
J5	MEM AD/WR 02	4					
J6	MEM AD/WR 02	4					
J7	MEM AD/WR 03	5					
J8	MEM AD/WR 03	5					
J9	MEM AD/WR 04	6					
J10	MEM AD/WR 04	6					
J11	MEM AD/WR 05	7					
J12	MEM AD/WR 05	7					
J13	MEM AD/WR 06	8					
J14	MEM AD/WR 06	8					
J15	MEM AD/WR 07	9					
J16	MEM AD/WR 07	9					
J17	MI PRESENT	28					
J18	WR RAM	30					
J19	ADRS COMPARE	33					
J20	CLR/TIMEOUT	39	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	NDRO ENBL	41	6.0	0.0	0.5	V	1-4
J22	INTERNAL ATR	46					
J23	CB BSY	47					
J24	CYCLE INIT	48					
J25	CB ATR	49					
J26	CB CORE	50					
J27	MEM AD/WR 15	52					
J28	MEM AD/WR 15	52					
J29	MEM AD/WR 15	52					
J30	MEM AD/WR 14	53					
J31	MEM AD/WR 14	53					
J32	MEM AD/WR 14	53					
J33	MEM AD/WR 13	54					
J34	MEM AD/WR 13	54					
J35	MEM AD/WR 13	54					
J36	MEM AD/WR 12	55					
J37	MEM AD/WR 12	55					
J38	MEM AD/WR 12	55					
J39	MEM AD/WR 11	56					
J40	MEM AD/WR 11	56	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	MEM AD/WR 11	56	6.0	0.0	0.5	V	1-4
J42	MEM AD/WR 10	57					
J43	MEM AD/WR 10	57					
J44	MEM AD/WR 10	57					
J45	MEM AD/WR 09	58					
J46	MEM AD/WR 09	58					
J47	MEM AD/WR 09	58					
J48	MEM AD/WR 08	59					
J49	MEM AD/WR 08	59					
J50	MEM AD/WR 08	59					
J51	RMW	69					
J52	DTR HOLDOFF	70					
J53	MEM ADRS 19	72					
J54	MEM ADRS 20	73					
J55	LATCH FCTN	78					
J56	DATA READY	79					
J57	CORE	82					
J58	ADRS CMPR 15	86					
J59	MEM ADRS 16	87					
J60	MEM ADRS 17	88	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J61	MEM ADRS 18	89	6.0	0.0	0.5	V	1-4
J62	MOD ENBL 0	91					
J63	MOD ENBL 1	92					
J64	MOD ENBL 2	93					
J65	MOD ENBL 3	94	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J65
Power supply current limit	1.68 A	NA	J1-J65
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J65
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J65 J1-J65
Output voltage measurement	NA	± 25 mV	J1-J65

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	MI PRESENT	28	5.5	0.0	0.5	mA	1-3
K2	ADRS COMPARE	33	5.5	0.0	0.25	mA	1-3
K3	CB CORE	50	2.4	-0.05	0.05	mA	1-3
K4	DATA READY	79	5.5	0.0	0.25	mA	1-3
Low level							
K5	CB CORE	50	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K5
Power supply current limit	1.68 A	NA	K1-K5
Forcing voltage,			
(Voh)	5.5 V	± 50 mV	K1,K2
(Voh)	2.4 V	± 50 mV	K3
(Voh)	5.5 V	± 50 mV	K4
(Vol)	0.4 V	± 50 mV	K5
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	K1-K5
low, (VL)	0.0 V	± 100 mV	K1-K5
Output current measurement	NA	± 25 uA	K1
	NA	± 13 uA	K2
	NA	± 5 uA	K3
	NA	± 13 uA	K4
	NA	± 5 uA	K5

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	CB AD/WR 00	11	-18.0	-1.2	0.0	V	1-3
L2	CB AD/WR 01	12					
L3	CB AD/WR 02	13					
L4	CB AD/WR 03	14					
L5	CB AD/WR 04	15					
L6	CB AD/WR 05	16					
L7	CB AD/WR 06	17					
L8	CB AD/WR 07	18					
L9	CB OR	19					
L10	LOOPBACK	20					
L11	FCTN 3	22					
L12	FCTN 0	23					
L13	CB NDRO SEL	24					
L14	CB MI ENBL	26					
L15	ABORT	27					
L16	CB WR ENBL	29					
L17	FIRST HALF	32					
L18	CLR BUS BSY	34	-18.0	-1.2	0.0	V	1-3
L19	STRAP BIT 15	36	-18.0	-1.5	0.0	V	1-3
L20	STRAP BIT 16	37	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	STRAP BIT 17	38	-18.0	-1.5	0.0	V	1-3
L22	STRAP BIT 18	42					
L23	STRAP BIT 19	43					
L24	STRAP BIT 20	44					
L25	STRAP BIT 21	45	-18.0	-1.5	0.0	V	1-3
L26	CB AD/WR 15	61	-18.0	-1.2	0.0	V	1-3
L27	CB AD/WR 14	62					
L28	CB AD/WR 13	63					
L29	CB AD/WR 12	64					
L30	CB AD/WR 11	65					
L31	CB AD/WR 10	66					
L32	CB AD/WR 09	67					
L33	CB AD/WR 08	68					
L34	TIMEOUT	71					
L35	CB CLK	74					
L36	FCTN 2	76					
L37	FCTN 1	77					
L38	MACLR	80	-18.0	-1.2	0.0	V	1-3
L39	NDRO STRAP	83	-18.0	-1.5	0.0	V	1-3
L40	RAM ADRS 0	84	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	RAM ADRS 3	85	-18.0	-1.2	0.0	V	1-3
L42	CB ADRS 16	95					
L43	CB ADRS 17	96					
L44	CB ADRS 18	97					
L45	CB ADRS 19	98					
L46	CB ADRS 20	99					
L47	CB ADRS 21	100	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L47
Power supply current limit	NA	NA	L1-L47
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L47
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L47 L1-L47
Input voltage measurement	 NA NA NA NA NA	 ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	 L1-L18 L19-L25 L26-L38 L39 L40-L47

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	5.7	10.7	uF	1-4
M2	PSFC	51	5.0	9.4	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 250 nF ± 220 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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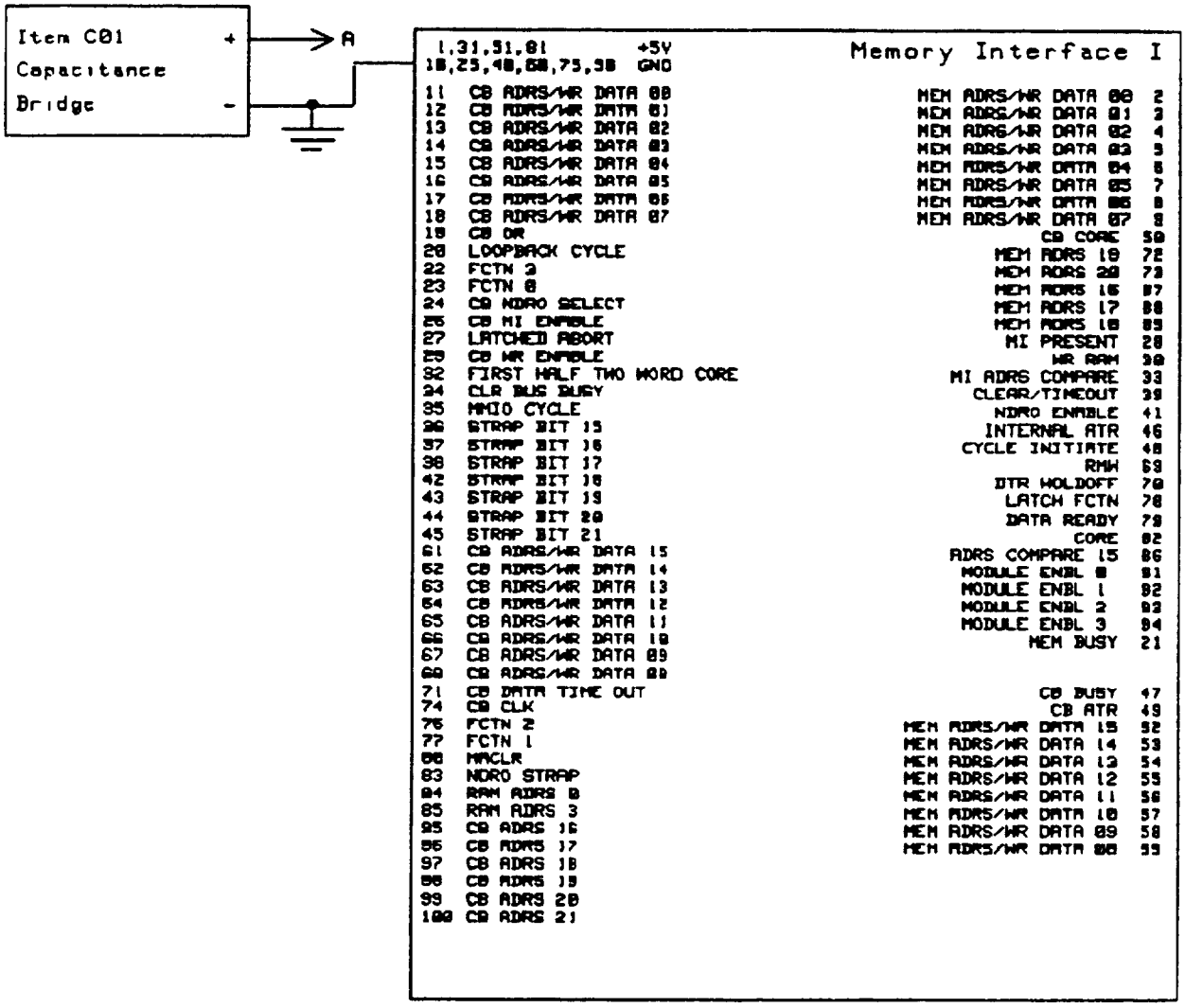


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	467	1121	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.68 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 32.7 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.68 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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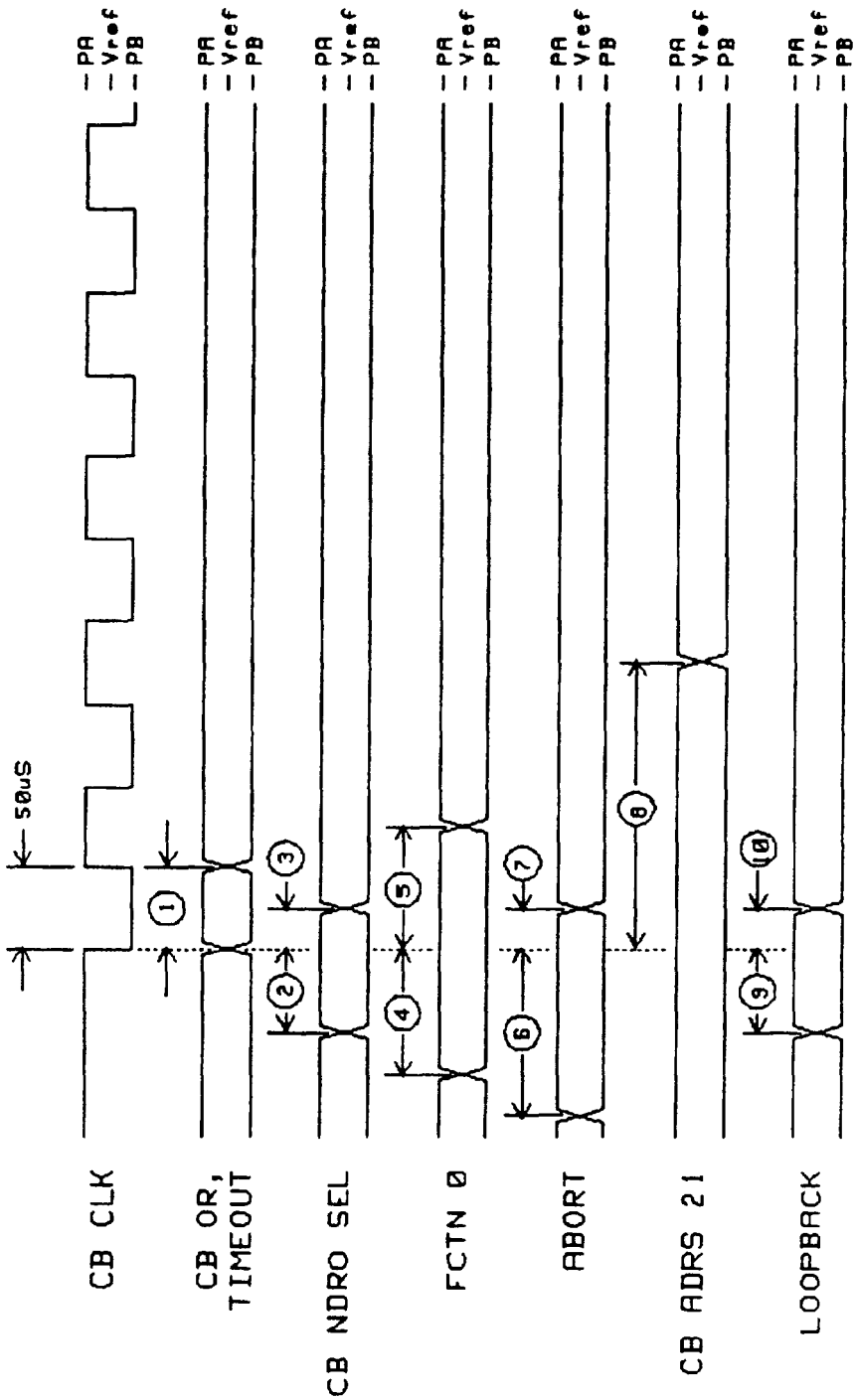


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) CB CLK to CB OR and TIMEOUT. Hold time: 11 nS minimum.

2. (2) CB NDRO SEL to CB CLK. Setup time: 12 nS minimum.

3. (3) CB CLK to CB NDRO SEL. Hold time: 8 nS minimum.

4. (4) FCTN 0 to CB CLK. Setup time: 29 nS minimum.

5. (5) CB CLK to FCTN 0. Hold time: 26 nS minimum.

6. (6) ABORT to CB CLK. Setup time: 33 nS minimum.

7. (7) CB CLK to ABORT. Hold time: 5 nS minimum.

8. (8) CB CLK to CB ADRS 21. Hold time: 65 nS minimum.

9. (9) LOOPBACK to CB CLK. Setup time: 23 nS minimum.

10. (10) CB CLK to LOOPBACK. Hold time: 6 nS minimum.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.68 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

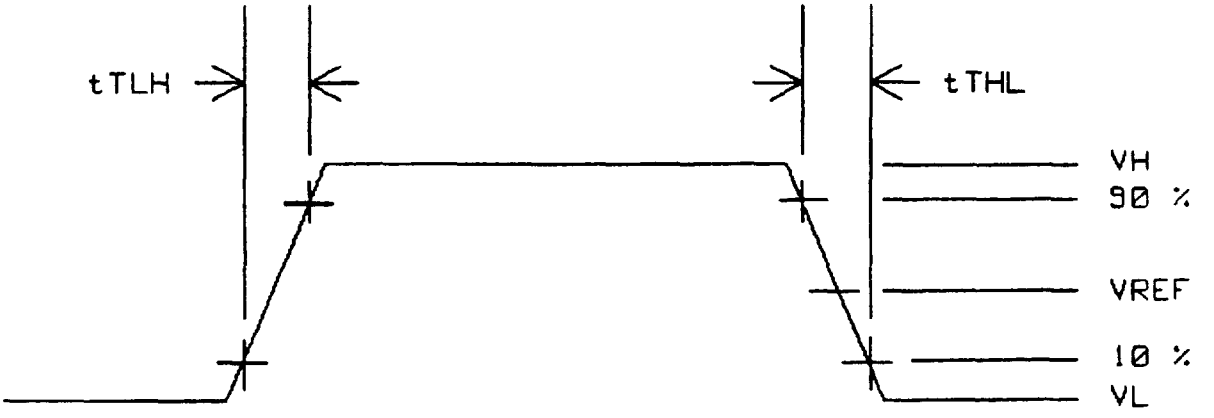
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLK to INTERNAL ATR	-	-	-	23	23	23	nS	12a
2	tpd	CB CLK to CB ATR	-	-	-	33	33	33	nS	12a
3	tpd	CB CLK to ADRS COMPARE	-	-	-	48	48	48	nS	12a
3	tpd	CB CLK to DTR HOLDOFF	-	-	-	48	48	48	nS	12a
4	tpd	CB CLK to MEM AD/WR 00	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 01	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 02	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 03	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 04	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 05	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 06	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 07	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 08	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 09	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 10	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 11	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 12	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 13	-	-	-	38	38	38	nS	12a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	CB CLK to MEM AD/WR 14	-	-	-	38	38	38	nS	12a
4	tpd	CB CLK to MEM AD/WR 15	-	-	-	38	38	38	nS	12a
5	tpd	RAM ADRS 0 to CORE	-	-	-	52	52	52	nS	12a
5	tpd	RAM ADRS 3 to CORE	-	-	-	52	52	52	nS	12a
6	th	CB CLK to CB OR	11	11	11	-	-	-	nS	12a
6	th	CB CLK to TIMEOUT	11	11	11	-	-	-	nS	12a
7	tsu	CB NDRO SEL to CB CLK	12	12	12	-	-	-	nS	12b
8	th	CB CLK to CB NDRO SEL	8	8	8	-	-	-	nS	12b
9	tsu	FCTN 0 to CB CLK	29	29	29	-	-	-	nS	12b
10	th	CB CLK to FCTN 0	26	26	26	-	-	-	nS	12b
11	tsu	ABORT to CB CLK	33	33	33	-	-	-	nS	12b
12	th	CB CLK to ABORT	5	5	5	-	-	-	nS	12b
13	th	CB CLK to CB ADRS 21	65	65	65	-	-	-	nS	12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
14	tsu	LOOPBACK to CB CLK	23	23	23	-	-	-	nS	12b
15	th	CB CLK to LOOPBACK	6	6	6	-	-	-	nS	12b

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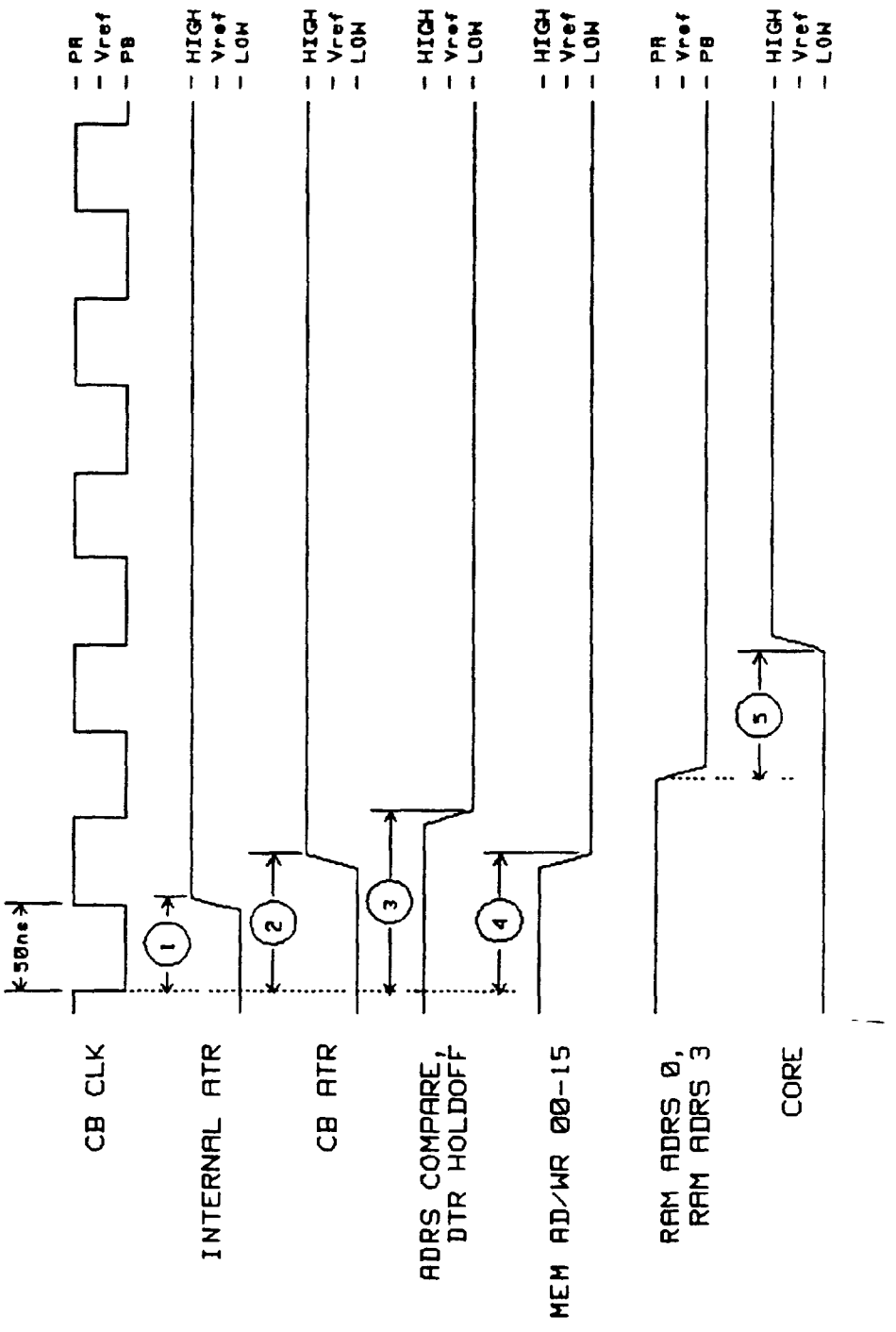


FIGURE 12a. Dynamic waveforms for table XVII.

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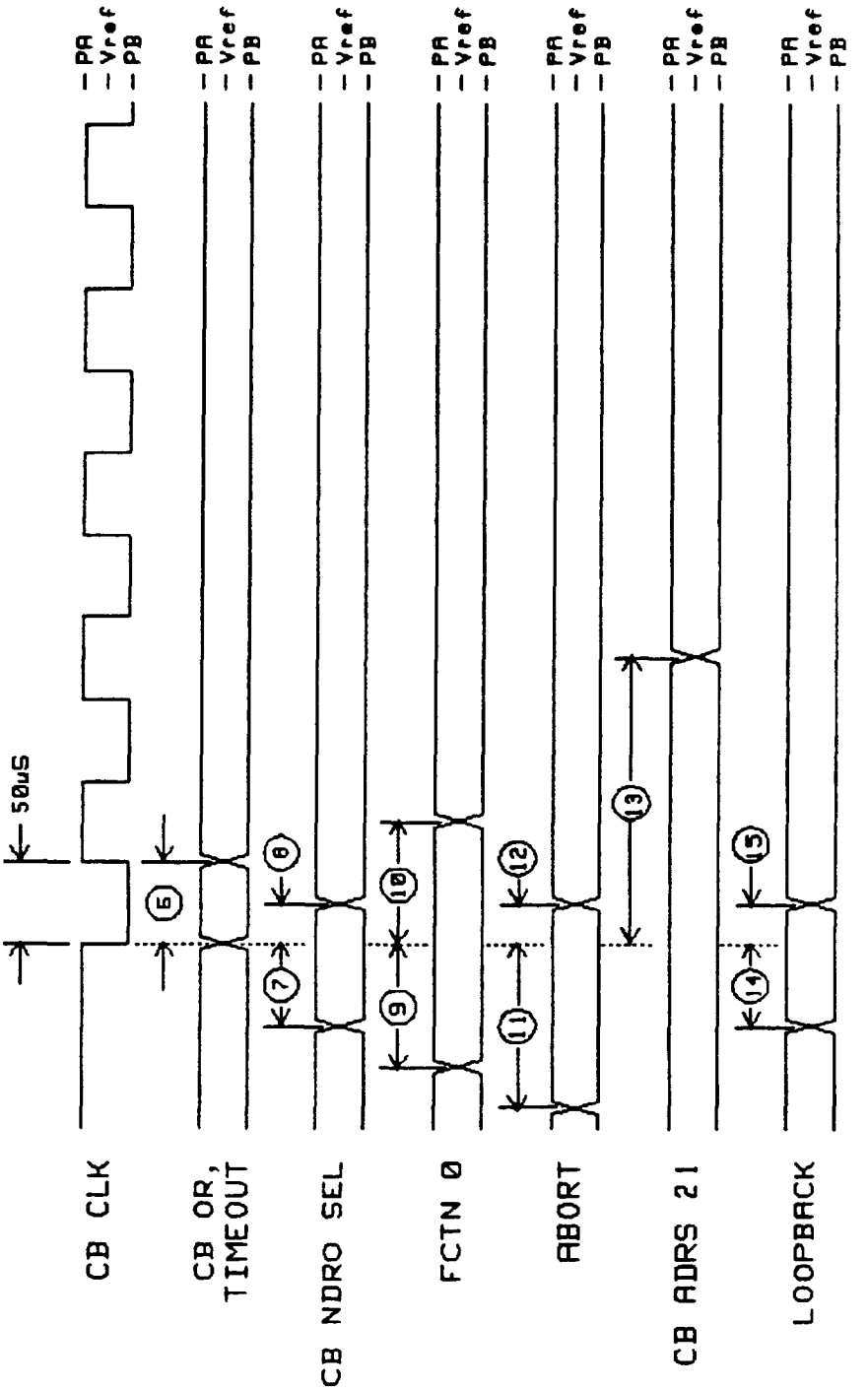


FIGURE 12b. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	CB AD/WR 00	11	2	50	pF	1-4
R2	CB AD/WR 01	12				
R3	CB AD/WR 02	13				
R4	CB AD/WR 03	14				
R5	CB AD/WR 04	15				
R6	CB AD/WR 05	16				
R7	CB AD/WR 06	17				
R8	CB AD/WR 07	18	2	50	pF	1-4
R9	CB OR	19	4	100	pF	1-4
R10	LOOPBACK	20	2	50	pF	1-4
R11	FCTN 3	22	6	125	pF	1-4
R12	FCTN 0	23	4	100	pF	1-4
R13	CB NDRO SEL	24	8	150	pF	1-4
R14	CB MI ENBL	26	2	75	pF	1-4
R15	ABORT	27	2	50	pF	1-4
R16	CB WR ENBL	29	2	50	pF	1-4
R17	FIRST HALF	32	8	125	pF	1-4
R18	CLR BUS BSY	34	4	75	pF	1-4
R19	MMIO CYCLE	35	6	75	pF	1-4
R20	STRAP BIT 15	36	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	STRAP BIT 16	37	2	75	pF	1-4
R22	STRAP BIT 17	38	2	75	pF	1-4
R23	STRAP BIT 18	42	2	50	pF	1-4
R24	STRAP BIT 19	43				
R25	STRAP BIT 20	44				
R26	STRAP BIT 21	45	2	50	pF	1-4
R27	CB AD/WR 15	61	4	100	pF	1-4
R28	CB AD/WR 14	62	2	50	pF	1-4
R29	CB AD/WR 13	63				
R30	CB AD/WR 12	64				
R31	CB AD/WR 11	65				
R32	CB AD/WR 10	66				
R33	CB AD/WR 09	67				
R34	CB AD/WR 08	68	2	50	pF	1-4
R35	TIMEOUT	71	2	75	pF	1-4
R36	CB CLK	74	2	50	pF	1-4
R37	FCTN 2	76	6	100	pF	1-4
R38	FCTN 1	77	2	125	pF	1-4
R39	MACLR	80	4	75	pF	1-4
R40	NDRO STRAP	83	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	RAM ADRS 0	84	2	75	pF	1-4
R42	RAM ADRS 3	85	2	75	pF	1-4
R43	CB ADRS 16	95	2	50	pF	1-4
R44	CB ADRS 17	96				
R45	CB ADRS 18	97				
R46	CB ADRS 19	98				
R47	CB ADRS 20	99				
R48	CB ADRS 21	100	2	50	pF	1-4
Outputs						
R49	MEM AD/WR 00	2	4	75	pF	1-4
R50	MEM AD/WR 01	3				
R51	MEM AD/WR 02	4				
R52	MEM AD/WR 03	5				
R53	MEM AD/WR 04	6				
R54	MEM AD/WR 05	7				
R55	MEM AD/WR 06	8				
R56	MEM AD/WR 07	9	4	75	pF	1-4
Bidirectional						
R57	CB ATR	49	8	125	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Output						
R58	CB CORE	50	2	75	pF	1-4
Bidirectionals						
R59	MEM AD/WR 15	52	8	100	pF	1-4
R60	MEM AD/WR 14	53	8	100	pF	1-4
R61	MEM AD/WR 13	54	8	125	pF	1-4
R62	MEM AD/WR 12	55	8	100	pF	1-4
R63	MEM AD/WR 11	56	8	150	pF	1-4
R64	MEM AD/WR 10	57	8	150	pF	1-4
R65	MEM AD/WR 09	58				
R66	MEM AD/WR 08	59	8	150	pF	1-4
Outputs						
R67	MEM ADRS 19	72	4	100	pF	1-4
R68	MEM ADRS 20	73	4	75	pF	1-4
R69	MEM ADRS 16	87	4	75	pF	1-4
R70	MEM ADRS 17	88	6	100	pF	1-4
R71	MEM ADRS 18	89	6	100	pF	1-4
Bidirectional						
R72	CB BSY	47	4	100	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R72
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R72
Capacitance measurement	NA	± 2.4 pF	R1-R8
	NA	± 4.8 pF	R9
	NA	± 2.4 pF	R10
	NA	± 5.95 pF	R11
	NA	± 4.8 pF	R12
	NA	± 7.1 pF	R13
	NA	± 3.65 pF	R14
	NA	± 2.4 pF	R15,R16
	NA	± 5.85 pF	R17
	NA	± 3.55 pF	R18
	NA	± 3.45 pF	R19
	NA	± 2.4 pF	R20
	NA	± 3.65 pF	R21,R22
	NA	± 2.4 pF	R23-R26
	NA	± 4.8 pF	R27
	NA	± 2.4 pF	R28-R34
	NA	± 3.65 pF	R35
	NA	± 2.4 pF	R36
	NA	± 4.7 pF	R37
	NA	± 6.15 pF	R38
	NA	± 3.55 pF	R39
	NA	± 2.4 pF	R40
	NA	± 3.65 pF	R41,R42
	NA	± 2.4 pF	R43-R48
	NA	± 3.55 pF	R49-R56
	NA	± 5.85 pF	R57
	NA	± 3.65 pF	R58
	NA	± 4.6 pF	R59,R60
	NA	± 5.85 pF	R61

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 4.6 pF	R62
	NA	± 7.1 pF	R63-R66
	NA	± 4.8 pF	R67
	NA	± 3.55 pF	R68,R69
	NA	± 4.7 pF	R70,R71
	NA	± 4.8 pF	R72

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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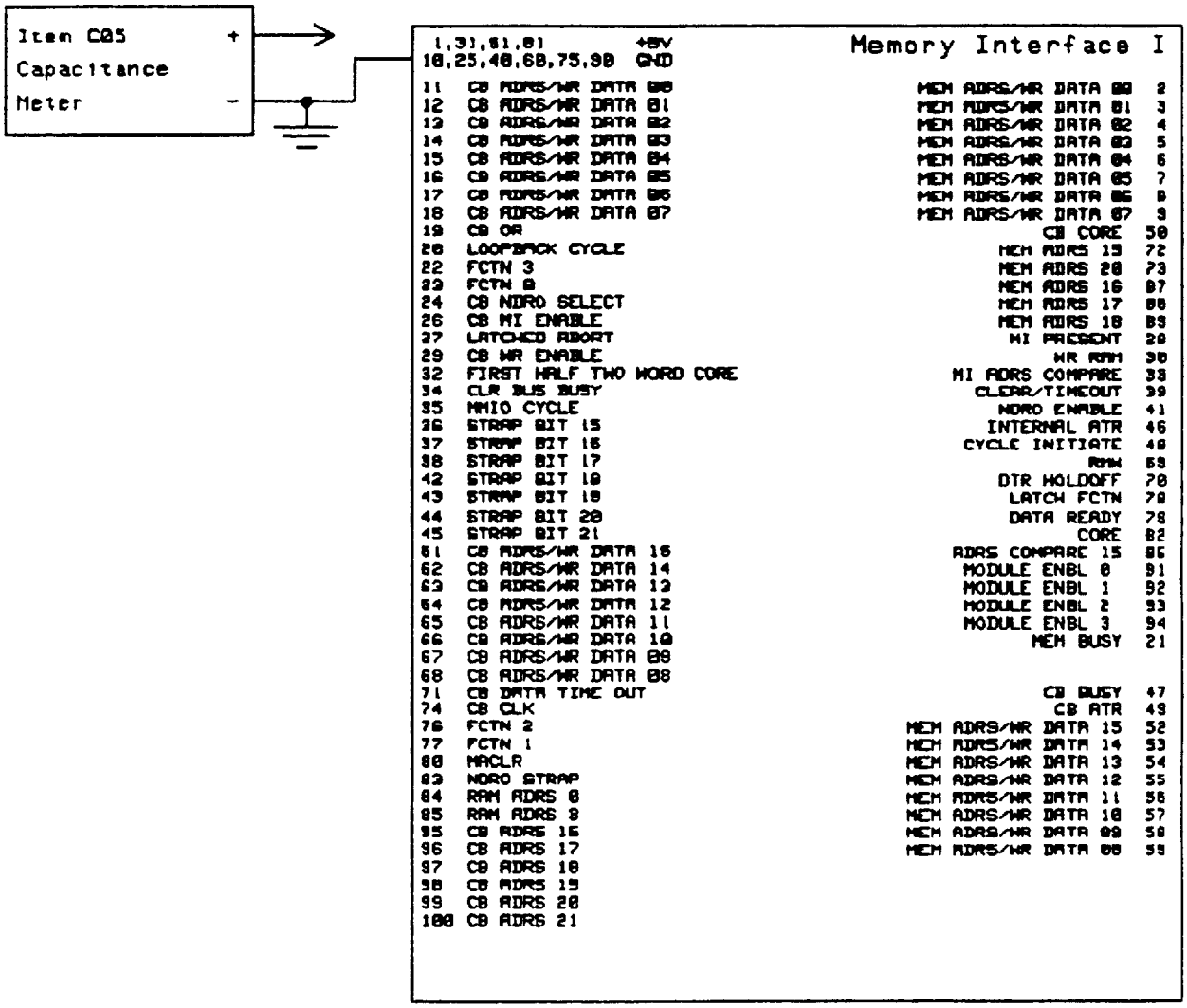


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.68 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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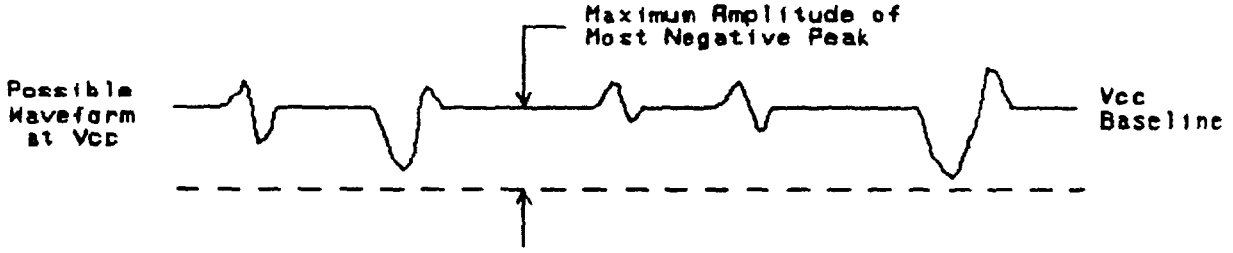
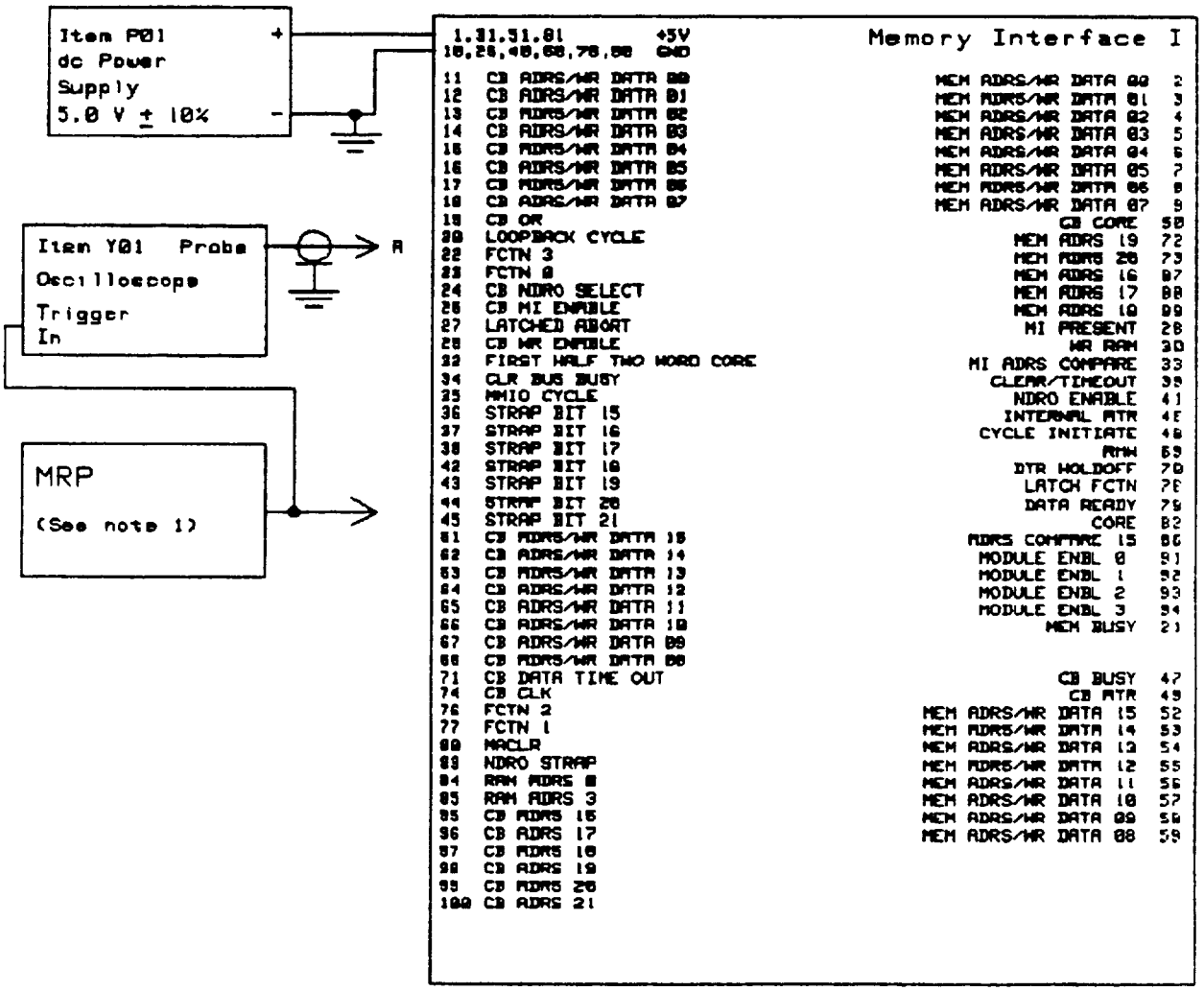


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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